

Sparse and Low-Rank Optimization for Dense Wireless Networks

Part I: Models

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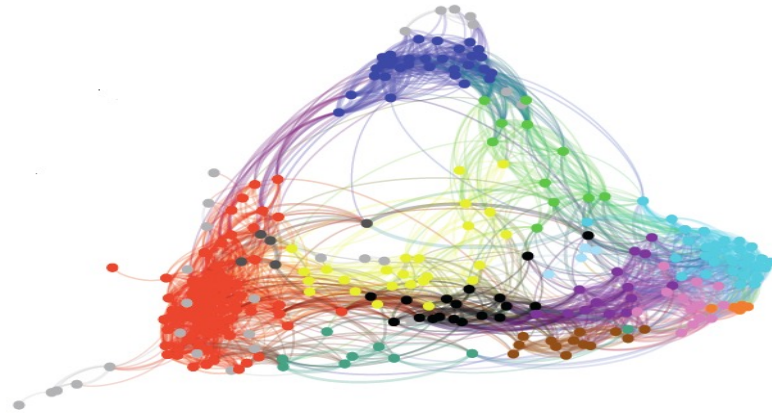


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Outline

- **Motivations**
- **Two Vignettes**
 - Structured Sparse Models
 - Group Sparse Beamforming for Network Power Minimization
 - Sparsity Control for Massive Device Connectivity
 - Generalized Low-Rank Models
 - Low-Rank Matrix Completion for Topological Interference Management
 - Extensions
- **Future Directions**

*Motivations: **Dense** Wireless Networks*



Ultra mobile broadband

- Era of mobile data traffic deluge



Source: Cisco VNI Mobile, 2017

18x
Over past 5 years

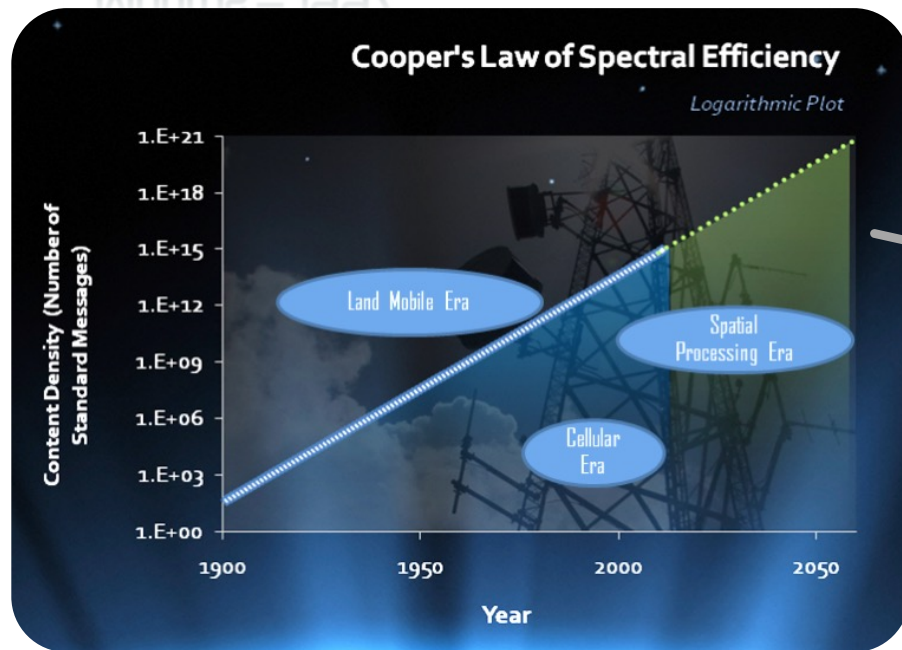


429 M
Mobile devices
added in 2016

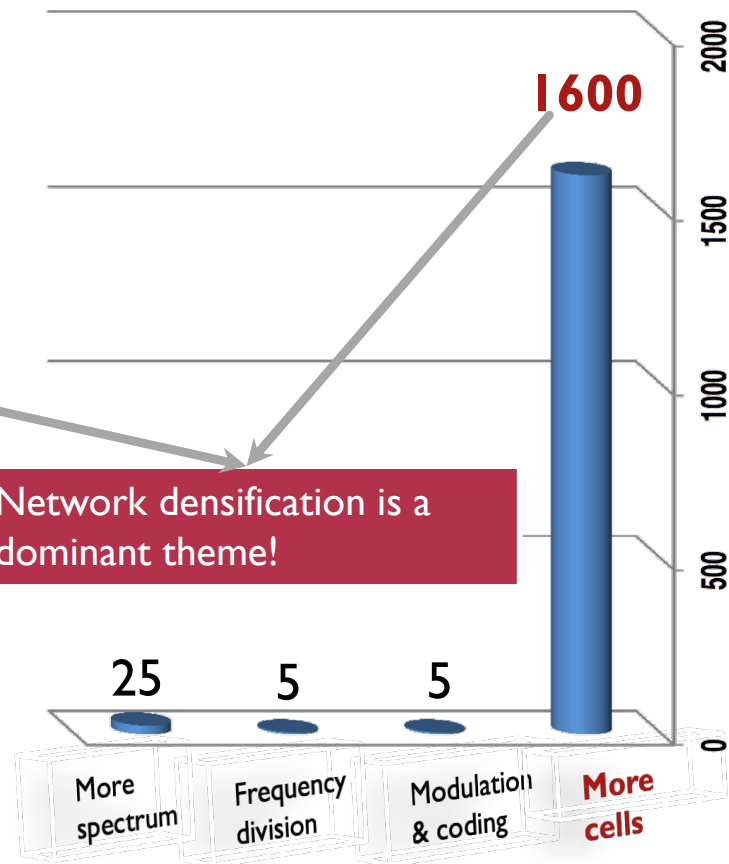
60%
in 2016

Cooper's Law

Marty Pens Cooper's Law: Data Over Usable Spectrum Doubles Every 30 Months – 1997

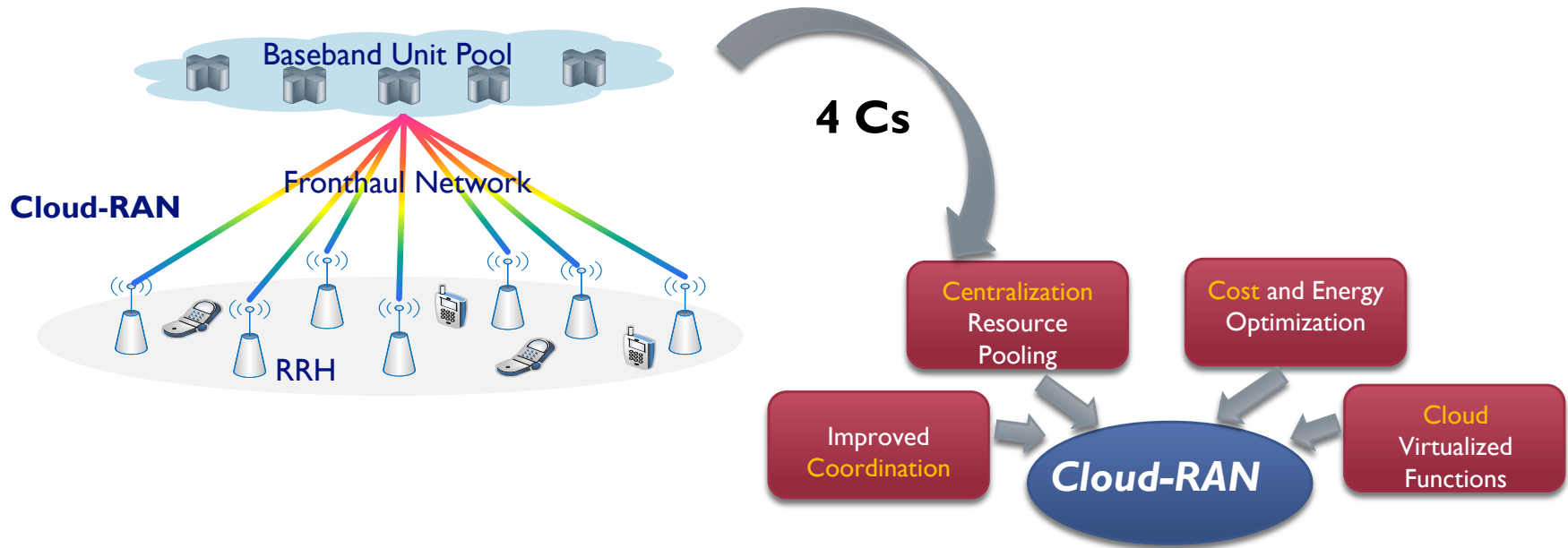


Factor of Capacity Increase since 1950



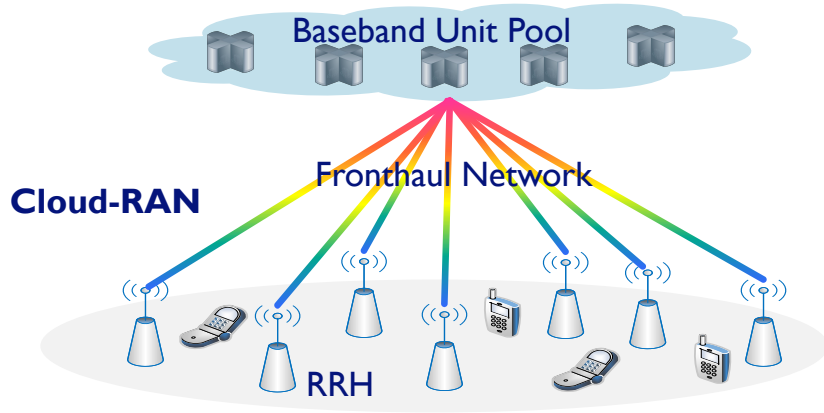
Solution: cloud radio access networks

- **Dense Cloud-RAN:** a cost-effective way for wireless network densification and cooperation (support Gbps communications)



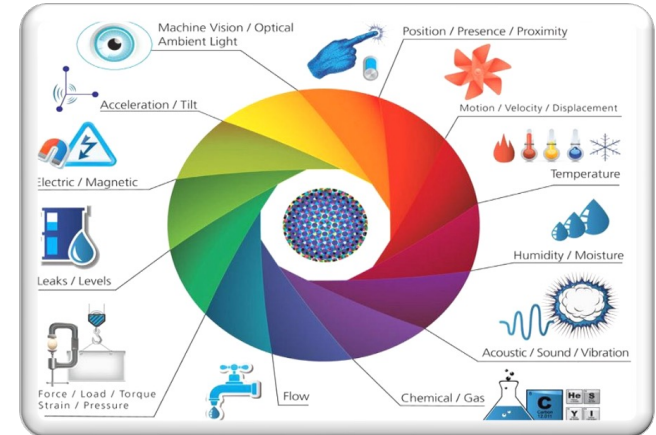
Dense Cloud-RAN

- Higher network capacity
 - Denser deployment
- Scalable connectivity
 - Flexible resource management
- Higher energy efficiency
 - Low-power RRH, flexible energy management
- Higher cost efficiency
 - Low-cost RRH, efficient resource utilization



Intelligent things for smart city

- A smart city highly depends on intelligent technology: connected sensors, intelligent devices and IoT networks become wholly integrated



Intelligent IoT



(internet of skills)

Tactile Internet

Internet of Things

Mobile Internet

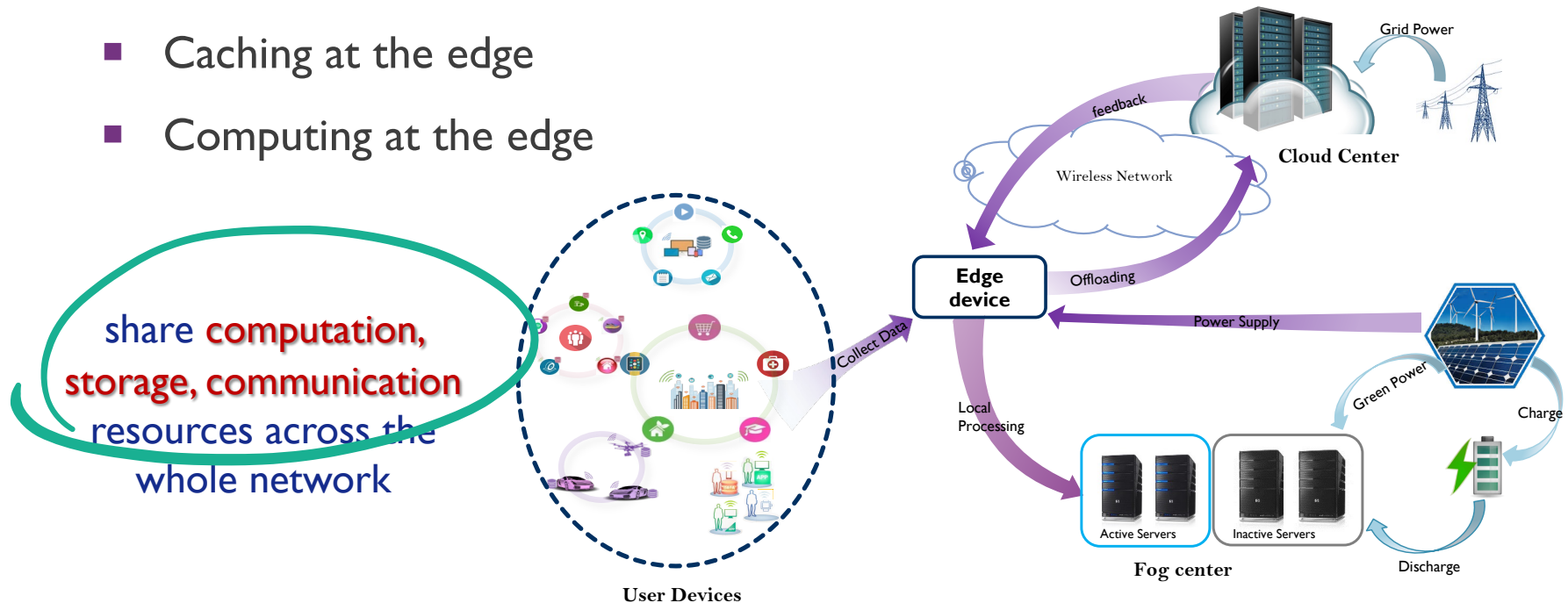
People to People



Fundamental shift: from content-delivery to skillset-delivery networks

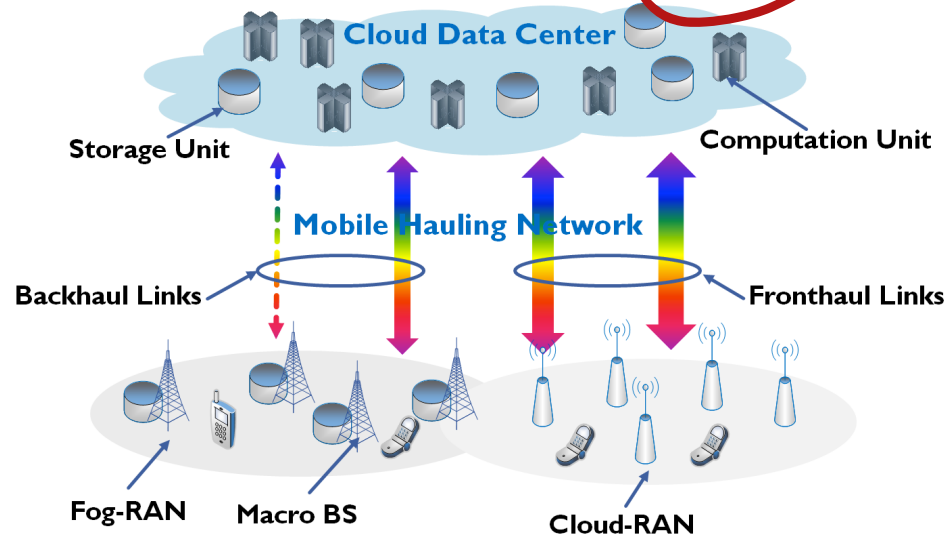
Solution: fog radio access networks

- **Dense Fog-RANs:** push computation and storage resources to network edge
 - Caching at the edge
 - Computing at the edge



A new paradigm for wireless networking

- **Goal:** support ultra-low latency, reliable, Gbps communications, massive device connectivity, massive data analytics, edge-AI...



Unique challenges

■ Networking issues:

- Huge network power consumption
- Massive device connectivity
- Severe network interference



Source: Alcatel-Lucent, 2013

■ Computing issues:

- Complicated (non-convex) problem structures
- Limited computational resources



Sparse and low-rank optimization

■ Examples

- **Compressed sensing/matrix completion:** Collect random measurements; reconstruct via optimization
- **Statistical machine learning:** Random data models; fit model via optimization

■ Motivations

- **Modeling flexibility:** Low-dimensional structures in high-dimensional data
- **Fundamental bounds:** Computational and statistical tradeoffs

Sparse and low-rank optimization

■ Emerging examples in wireless

- **Structured sparse models:** Group sparse beamforming, message sharing, user admission control, massive device connectivity...
- **Generalized low-rank models:** Topological interference management, mobile edge caching, wireless distributed computing, index coding...

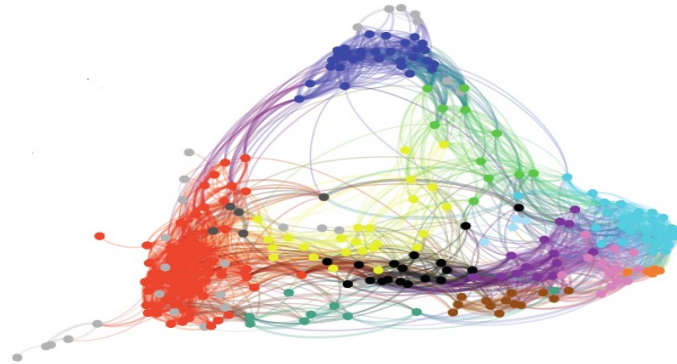
■ Motivations

- **Modeling flexibility:** Structured models in dense and complex networks
- **Computational scalability:** Convex optimization, manifold optimization...
- **Theoretical guarantees:** Convex geometry, differential geometry...

Vignette A: *Structured Sparse Models*

Case I: Group Sparse Beamforming for Network Power Minimization

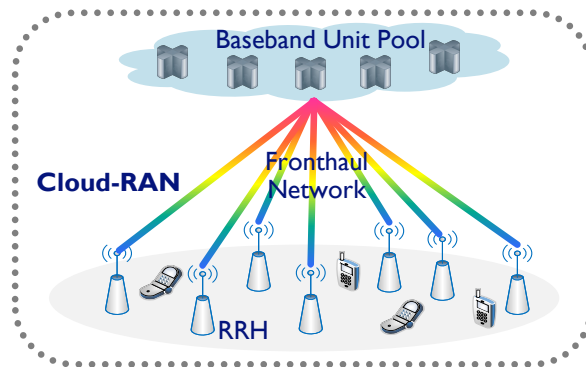
Case II: Sparsity Control for Massive Device Connectivity



Case I: *Group Sparse Beamforming* for *Network Power Minimization*

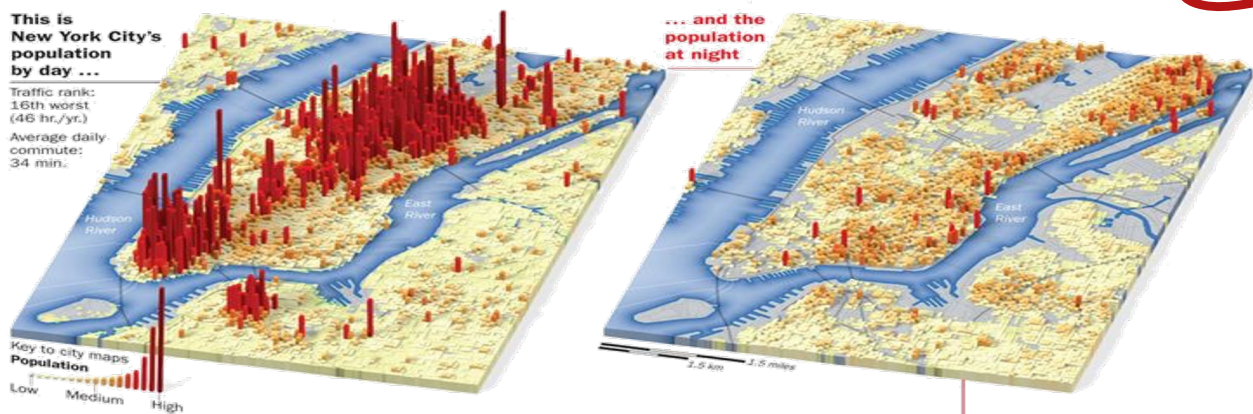
Network power consumption

- **Goal:** Design a green dense Cloud-RAN
- **Prior works:** Physical-layer *transmit power consumption*
 - Wireless power control: [Chiang, et al., FT 08], [Qian, et al., TWC 09], [Sorooshyari, et al., TON 12], ...
 - Transmit beamforming: [Sidiropoulos and Luo, TSP 2006], [Yu and Lan, TSP 07], [Gershman, et al., SPMag 10], ...
- **Unique challenge:**
 - Network power consumption:
 - Radio access units, fronthaul links, etc.



Network adaptation

- **Goal:** Provide a holistic approach to minimize network power consumption (including RRHs, fronthaul links, etc.)
- **Key observation:** Spatial and temporal mobile data traffic variation



Network adaptation: adaptively switch off network entities to save power

System model

- The received signal at the k -th MU is given by

$$\underline{y_k} = \sum_{l=1}^L \mathbf{h}_{kl}^H \underline{\mathbf{v}_{lk}} s_k + \sum_{i \neq k} \sum_{l=1}^L \mathbf{h}_{kl}^H \mathbf{v}_{li} s_i + n_k, \forall k$$

desired signal interference noise

$$S_k \sim \mathcal{CN}(0, 1)$$

$$P_l = E[\|X_l\|_2^2]$$

$$X_l = \sum_{k=1}^K \mathbf{v}_{lk} s_k$$

- $\mathbf{h}_{kl} \in \mathbb{C}^{N_l}$: channel propagation between MU k and RRH l
- $\mathbf{v}_{lk} \in \mathbb{C}^{N_l}$: transmit beamforming vector from the l -th RRH to k -th MU
 (pre-coding)
- Per-RRH transmit power constraint: $\mathcal{C} = \left\{ \mathbf{v}_{lk} : \sum_{k=1}^K \|\mathbf{v}_{lk}\|_2^2 \leq P_l, \forall l \right\}$

- The signal-to-interference-plus-noise ratio (SINR) for MU

$$\text{SINR}_k = \frac{|\mathbf{h}_k^H \mathbf{v}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} \geq \gamma_k, \forall k.$$

$$R_k = \log_2(1 + \text{SINR}_k)$$

$$\mathbf{h}_k \triangleq [\mathbf{h}_{k1}^T, \dots, \mathbf{h}_{kL}^T]^T \in \mathbb{C}^N$$

$$\mathbf{v}_k \triangleq [\mathbf{v}_{1k}^T, \mathbf{v}_{2k}^T, \dots, \mathbf{v}_{Lk}^T]^T \in \mathbb{C}^N$$

$$N = \sum_{l=1}^L N_l$$

Network power consumption

- **Continuous function:** Transmit power consumption

$$T(\mathbf{v}) = \sum_{l=1}^L \sum_{k=1}^K \frac{1}{\eta_l} \|\mathbf{v}_{lk}\|_2^2$$

- $\eta_l > 0$: Drain inefficiency coefficient of the radio frequency power amplifier

- **Combinatorial function:** Relative fronthaul link power consumption

$$F(\mathcal{T}(\mathbf{v})) = \sum_{l=1}^L \underline{P_l^c} I(\text{Supp}(\mathbf{v}) \cap \mathcal{V}_\ell \neq \emptyset)$$

- $\mathcal{V}_l = \{K \sum_{i=1}^{l-1} N_i + 1, \dots, K \sum_{i=1}^l N_i\}, \forall l$: a partition of $\mathcal{V} = \{1, \dots, KN\}$
- $P_l^c \geq 0$: relative fronthaul link power consumption, i.e., the static power saving when both the fronthaul link and the corresponding RRH are switched off

- Aggregative beamformers: $\mathbf{v} = [\underbrace{\mathbf{v}_{11}^T, \dots, \mathbf{v}_{1K}^T}_{\tilde{\mathbf{v}}_1^T}, \dots, \underbrace{\mathbf{v}_{L1}^T, \dots, \mathbf{v}_{LK}^T}_{\tilde{\mathbf{v}}_L^T}]^T$
group sparse

Problem formulation

- **Goal:** Minimize network power consumption in Cloud-RAN

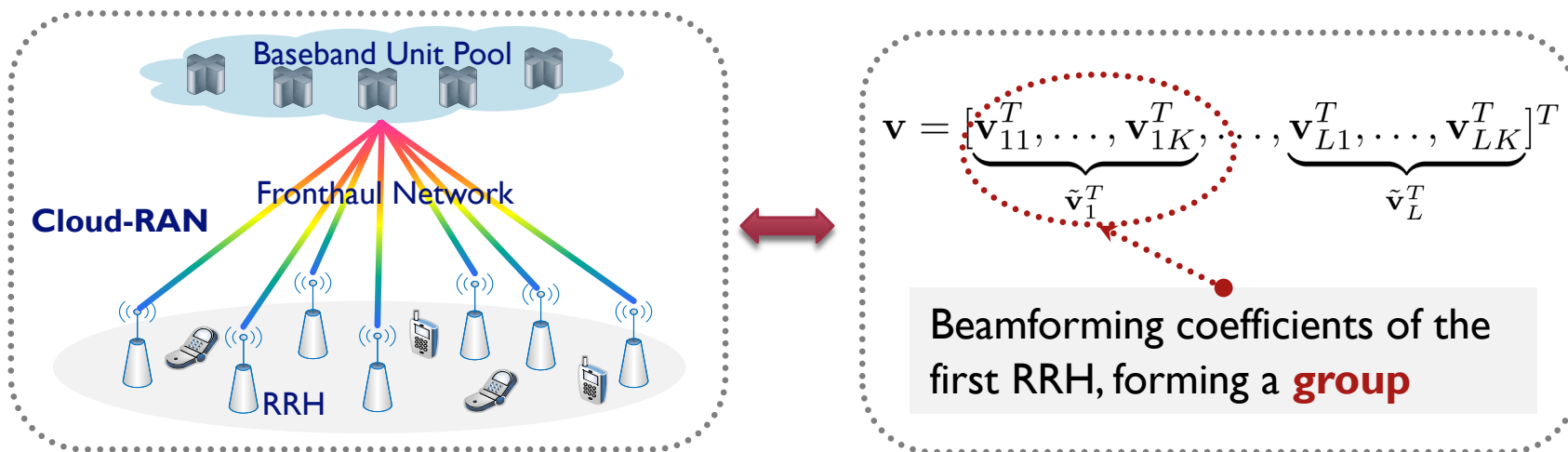
$$\underset{\mathbf{v} \in \mathcal{C}}{\text{minimize}} \quad F(\text{Supp}(\mathbf{v})) + T(\mathbf{v}) \quad \text{combinatorial composite function}$$

$$\text{subject to} \quad \frac{|\mathbf{h}_k^H \mathbf{v}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} \geq \gamma_k, \forall k.$$

- Simultaneously control both the combinatorial function F and the continuous function T
- **Prior algorithms:** heuristic or computationally expensive: [Philipp, et. al, TSP 13], [Luo, et. al, JSAC 13], [Quek, et. al, TWC 13],...

Finding structured solutions

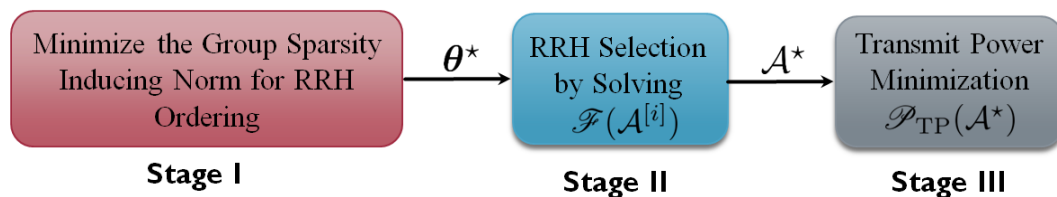
- **Proposal:** group sparse beamforming framework



- Switch off the l -th RRH $\rightarrow \tilde{\mathbf{v}}_l = \mathbf{0}$, i.e., **group sparsity structure** in $\mathbf{v}$

Group sparse beamforming algorithm

- Adaptive RRH selection: Switch off the RRHs with smallest coefficients in the aggregative beamformers



- Stage I:** The *tightest* convex positively homogeneous lower bound of the combinatorial composite objective function (*network power*)

$$\Omega(\mathbf{v}) = 2 \sum_{l=1}^L \sqrt{\frac{P_l^c}{\eta_l}} \|\tilde{\mathbf{v}}_l\|_2$$

mixed ℓ_1/ℓ_2 -norm

induce group sparsity

$$\theta^* = [\|\tilde{\mathbf{v}}_l^*\|_2]_{l=1}^L$$

ordering results

$$\mathbf{x} = [\|\tilde{\mathbf{v}}_1\|_2, \|\tilde{\mathbf{v}}_2\|_2, \dots, \|\tilde{\mathbf{v}}_L\|_2] \Rightarrow \|\mathbf{x}\|_1 = \sum_{l=1}^L \|\tilde{\mathbf{v}}_l\|_2$$

l1/l2 norm

Group sparse beamforming algorithm

- **Stage II:** Find the optimal active RRHs via solving the a sequence of the following feasibility detection problems (bi-section search approach)

$$\begin{aligned} \mathcal{F}(\mathcal{A}^{[i]}) : \text{find } & \mathbf{v}_1, \dots, \mathbf{v}_K & \mathbf{v}_k &= [\mathbf{v}_{lk}] \in \mathbb{C}^{|\mathcal{A}|N} \\ \text{subject to } & \frac{|\mathbf{h}_k^H \mathbf{v}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} \geq \gamma_k, \forall k & \mathbf{h}_k &= [\mathbf{h}_{lk}] \in \mathbb{C}^{|\mathcal{A}|N} \end{aligned}$$

- **Stage III:** Transmit power minimization via coordinated beamforming

$$\begin{aligned} \text{minimize}_{\mathbf{v} \in \mathcal{C}} \quad & T(\mathbf{v}; \mathcal{A}^*) \quad \text{Active RRH set} \\ \text{subject to } & \frac{|\mathbf{h}_k^H \mathbf{v}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} \geq \gamma_k, \forall k. & \mathbf{v}_k &= [\mathbf{v}_{lk}] \in \mathbb{C}^{|\mathcal{A}^*|N} \\ & & \mathbf{h}_k &= [\mathbf{h}_{lk}] \in \mathbb{C}^{|\mathcal{A}^*|N} \end{aligned}$$

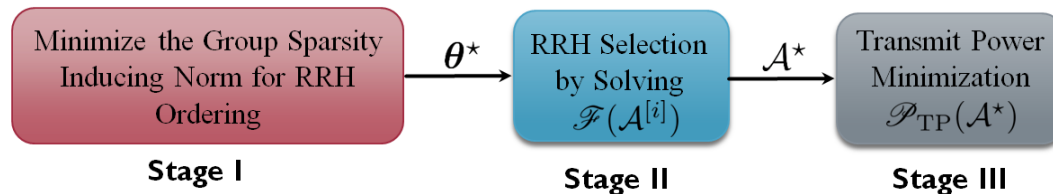
Summary of group sparse beamforming

- SINR constraints can be reformulated as second-order cone constraints

$$f_k(\mathbf{v}) = \sqrt{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} - \frac{1}{\sqrt{\gamma_k}} \Re(\mathbf{h}_k^H \mathbf{v}_k) \leq 0, \forall k. \quad \text{convex}$$

- **Key observation:** phases of $\mathbf{v}_k$'s do not change objective and constraints

- Group sparse beamforming via convex programming



- Stage I: Group sparsity inducing via solving one convex program
- Stage II: **A sequence of convex feasibility problems** need to be solved
- Stage III: Coordinated beamforming via solving one convex program

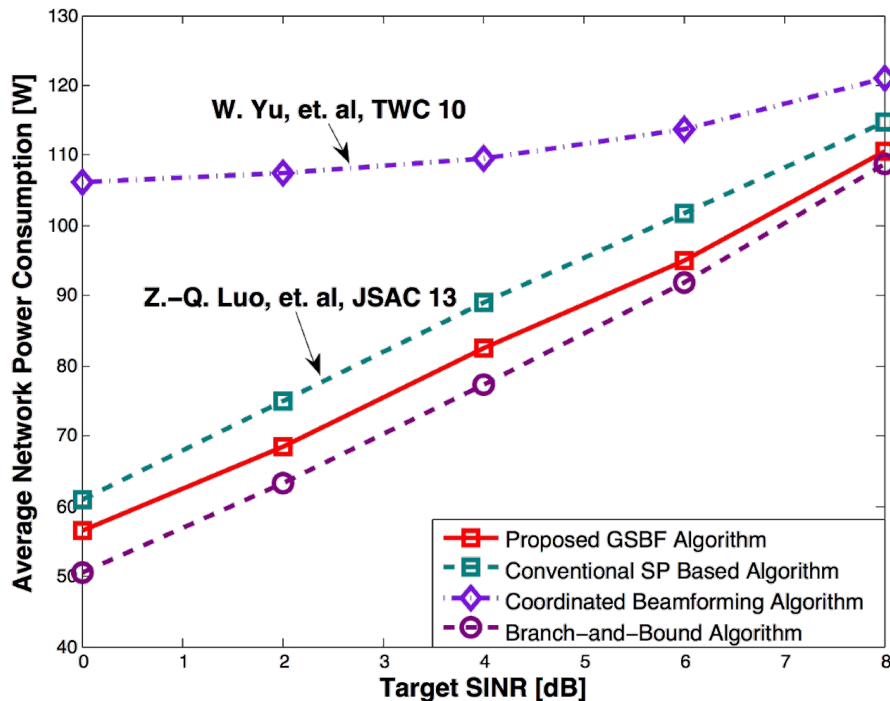
$$SNR_k = \frac{|h_k^H v_k|^2}{\sum_{i \neq k} |h_k^H v_i|^2 + b_k^2} \geq \gamma_k$$



$$\sqrt{\sum_{i \neq k} |h_k^H v_i|^2 + b_k^2} \leq \frac{1}{\sqrt{\gamma_k}} |h_k^H v_k|$$

The power of group sparse beamforming

- Group sparse beamforming for green Cloud-RAN (10 RRHs, 15 MUs)



Conclusions:

- 1) Enabling flexible network adaptation;
- 2) Offering efficient algorithm design via convex programming
- 3) Empowering wide applications

*Case II: **Sparsity Control** for **Massive Device** **Connectivity***

Motivation

- Downlink transmission with massive devices: **user admission control**
- Uplink machine-type communication (e.g., IoT devices) with sporadic traffic: **massive device connectivity**



Sporadic traffic: only a small fraction of potentially large number of devices are active

User admission control in downlink

- Coordinated beamforming for transmission power minimization

$$\begin{aligned} & \underset{\mathbf{v} \in \mathcal{C}}{\text{minimize}} && \|\mathbf{v}\|_2^2 \\ & \text{subject to} && \frac{|\mathbf{h}_k^H \mathbf{v}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} \geq \gamma_k, \forall k. \end{aligned}$$

- SINR constraints can be reformulated as second-order cone constraints

$$f_k(\mathbf{v}) = \sqrt{\sum_{i \neq k} |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_k^2} - \frac{1}{\sqrt{\gamma_k}} \Re(\mathbf{h}_k^H \mathbf{v}_k) \leq 0, \forall k.$$

- **Key observation:** phases of $\mathbf{v}_k$'s do not change objective and constraints

Infeasibility

- Set of convex inequalities:

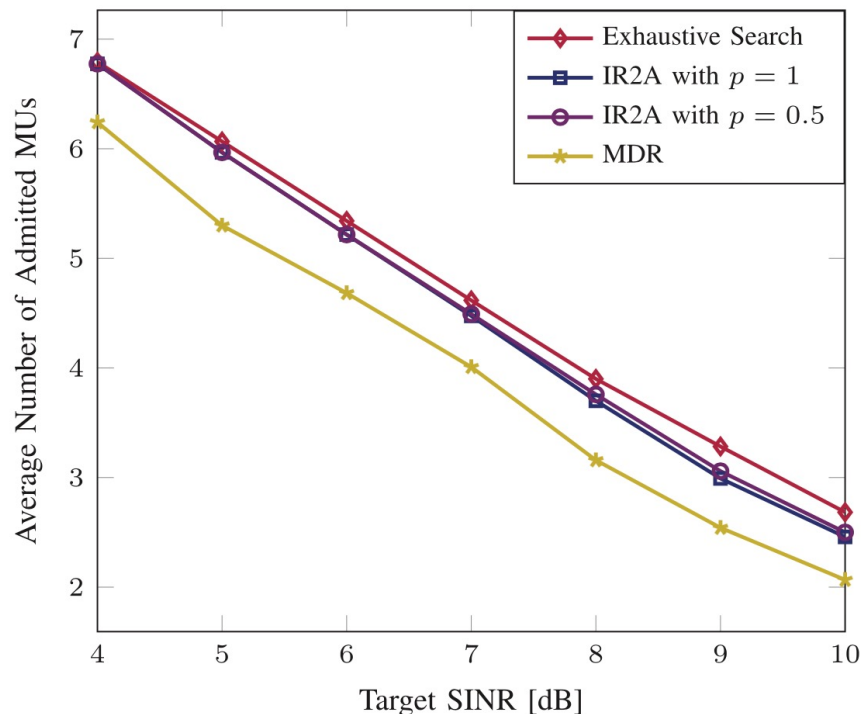
$$f_1(\mathbf{v}) \leq 0, \dots, f_m(\mathbf{v}) \leq 0, \quad \mathbf{v} \in \mathcal{C}$$

- Power minimization problem is generally **infeasible**: large number of users, unfavorable channel conditions, high data rate requirements,...
- **Goal**: Maximize the user capacity, i.e., the number of admitted users
- **Solution**: Choose $\mathbf{v}$ to minimize the number of violated inequalities

$$\begin{aligned} & \underset{\mathbf{v} \in \mathcal{C}, \mathbf{s}}{\text{minimize}} && \|\mathbf{s}\|_0 \\ & \text{subject to} && f_i(\mathbf{v}) \leq s_i, i = 1, \dots, m \\ & && s_i \geq 0, i = 1, \dots, m \end{aligned}$$

Sparse optimization for user admission control

- Average number of admitted mobile users versus target SINR



IR2A: iterative reweighted ℓ_2 -algorithm

$$\|s\|_0 \rightarrow \|s\|_p (0 < p < 1)$$

MDR: membership deflation by convex relaxation ($\|s\|_0 \rightarrow \|s\|_1$)

Massive device connectivity in uplink

- Cellular network with massive number of devices
 - Single-cell uplink with a BS with M antennas; Block-fading channel with coherence time T ; Total N single-antenna devices, $\mathcal{S} \subset \{1, 2, \dots, N\}$ devices are active (sporadic traffic)

$$\mathbf{y}(\ell) = \sum_{i \in \mathcal{S}} \mathbf{h}_i q_i(\ell) + \mathbf{n}(\ell), \ell = 1, \dots, L$$

- Define diagonal activity matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ with $|\mathcal{S}|$ non-zero diagonals

$$\mathbf{Y} = \mathbf{Q} \mathbf{A} \mathbf{H} + \mathbf{N}$$

- $\mathbf{Y} = [\mathbf{y}(1), \dots, \mathbf{y}(L)]^T \in \mathbb{C}^{L \times M}$ denotes the received signal across M antennas
- $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_N]^T \in \mathbb{C}^{N \times M}$: channel matrix from all devices to the BS
- $\mathbf{Q} = [\mathbf{q}(1), \dots, \mathbf{q}(L)]^T \in \mathbb{C}^{L \times N}$: known transmit pilot matrix from devices

Challenges of massive connectivity

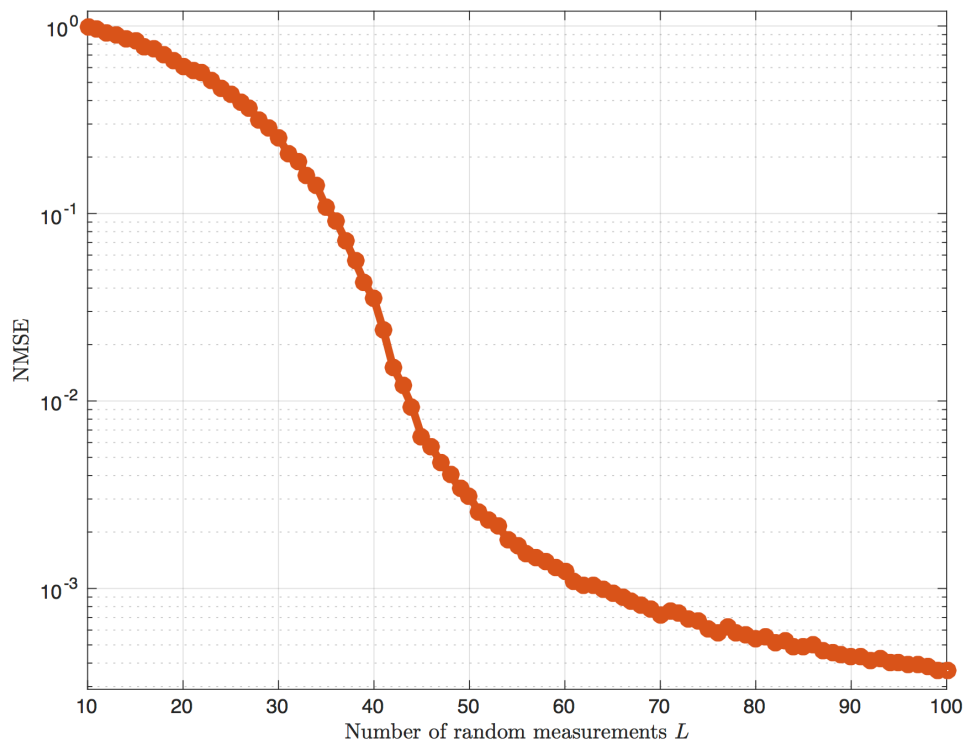
- **Sporadic traffic**
 - User activity detection is a key requirement
 - Massive number of devices means pilot sequences cannot be orthogonal
 - Device identification is a sparse optimization problem
- Prior work on compressed sensing for massive connectivity: 1) Without channel estimation [Zhang-Luo-Guo'13]; 2) Joint user activity detection and channel estimation [Xu-Rao-Lau'15]; 3) Approximate Message Passing (AMP) [Wei'16]
- **Proposal:** User activity detection and channel estimation based on the compressed sensing techniques (**without channel distribution prior**)

Group sparsity estimation

- Let $\Theta^{\natural} = \mathbf{A}\mathbf{H} \in \mathbb{C}^{N \times M}$ (unknown): group sparsity in rows $\theta^{[i]}$ of matrix $\Theta^{\natural}$
 - Simultaneous user activity detection and channel estimation
- Let $\mathbf{Q} \in \mathbb{C}^{L \times N}$ be a known measurement operator (pilot matrix)
- **Observe** $\mathbf{Y} = \mathbf{Q}\Theta^{\natural} + \mathbf{N}$
- Find estimate $\hat{\Theta}$ by solving a **convex program**
$$\underset{\Theta \in \mathbb{C}^{N \times M}}{\text{minimize}} \quad f(\Theta) \quad \text{subject to} \quad \|\mathbf{Y} - \mathbf{Q}\Theta\|_F \leq \epsilon$$
 - $f(\Theta) = \sum_{i=1}^N \|\theta^{[i]}\|_2$ is mixed ℓ_1/ℓ_2 -norm to reflect group sparsity structure
- **Hope:** $\hat{\Theta} = \Theta^{\natural}$

Sparse estimation for massive connectivity

- Normalized MSE versus pilot matrix length L



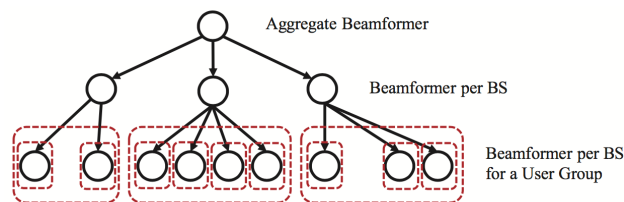
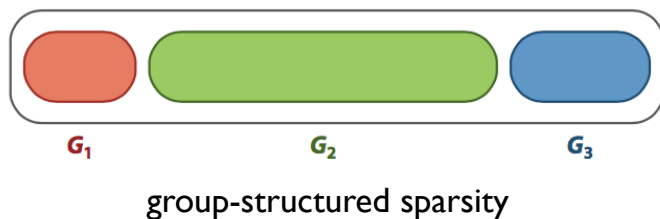
$$N = 100, M = 2, |\mathcal{S}| = 20$$

Summary: structured sparse models

- Generalized structured sparse optimization for dense networks

$$\underset{\mathbf{z} \in \mathbb{C}^n}{\text{minimize}} \quad f(\mathbf{z}) := \alpha f_1(\text{Supp}(\mathbf{z})) + \beta f_2(\mathbf{z}) \quad \text{subject to} \quad \mathbf{z} \in \mathcal{C}$$

- $\text{Supp}(\mathbf{z})$ is the index set of non-zero coefficients of a vector $\mathbf{z}$
- f_1 : combinatorial positive-valued set-function to control sparsity in $\mathbf{z}$
- f_2 : continuous convex function to represent the system performance
- $\mathcal{C}$: model system constraints, e.g., QoS constraints



layered group
sparsity

Vignette B: *Generalized Low-Rank Models*

Case I: Low-Rank Matrix Completion for Topological Interference Management

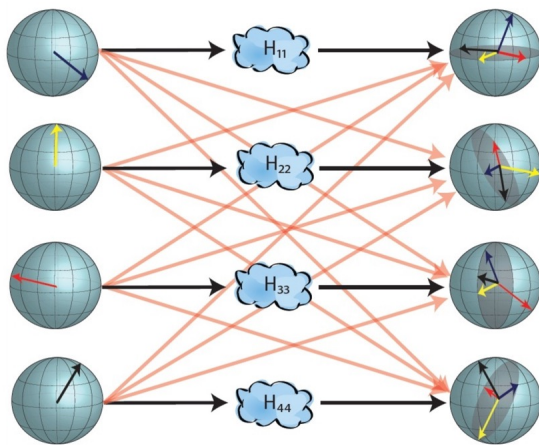
Case II: Extensions to Mobile Edge Caching, Distributed Computing and Index Coding

1		0	0	
	1	0	0	
0		1		0
0			1	0
	0			1

*Case I: **Topological** Interference Management*

Interference channel

- **Channel model:** $y_i = h_{ii}x_i + \sum_{j \neq i} h_{ij}x_j + z_j, i = 1, \dots, n$



capacity is unknown

- **Degrees-of-freedom:** simplify the analysis; lead to physical insights

$$\text{DoF} = \lim_{\text{SNR} \rightarrow \infty} \frac{C(\text{SNR})}{\log(\text{SNR})}$$

Interference alignment

- Assume the channel coefficients change over time:

$$y_i(t) = h_{ii}(t)x_i(t) + \sum_{j \neq i} h_{ij}(t)x_j(t) + z_j(t)$$

- Consider T channel uses: $X_i = [x_i(1), \dots, x_i(T)]^\dagger$, $Z_i = [z_i(1), \dots, z_i(T)]^\dagger$

$$Y_i = H_{ii}X_i + \sum_{j \neq i} H_{ij}X_j + Z_i \quad H_{ij} = \begin{bmatrix} h_{ij}(1) & & \\ & \ddots & \\ & & h_{ij}(T) \end{bmatrix}$$

- Each transmitter j sends m information symbols S_j across T channel uses

$$\sum_{j \neq i} H_{ij} V_j S_j \quad V_j \in \mathbb{C}^{T \times m} \text{ represents the precoding matrix}$$

- The i -th interference term $X_j = V_j S_j$ lives in the range space of the matrix

$$[H_{i1}V_1 \ \dots \ H_{i,i-1}V_{i-1} \ H_{i,i+1}V_{i+1} \ \dots \ H_{in}V_n]_{T \times (n-1)m}$$

Interference alignment

- **Interference alignment condition:** find precoding matrices and decoding matrices $V_j \in \mathbb{C}^{T \times m}, U_i \in \mathbb{C}^{m \times T}$ such that

$$\text{rank}(U_i H_{ii} V_i) = m, \forall i = 1, \dots, n$$

$$U_i [H_{i1} V_1 \ \dots \ H_{i,i-1} V_{i-1} \ H_{i,i+1} V_{i+1} \ \dots \ H_{in} V_n] = 0, \forall i = 1, \dots, n$$

- Each user can send m symbols: interference free across T channel uses

$$\text{DoF} = m$$

- **Intuition:** The interference has aligned onto a $T - m$ dimensional subspace at each receiver.

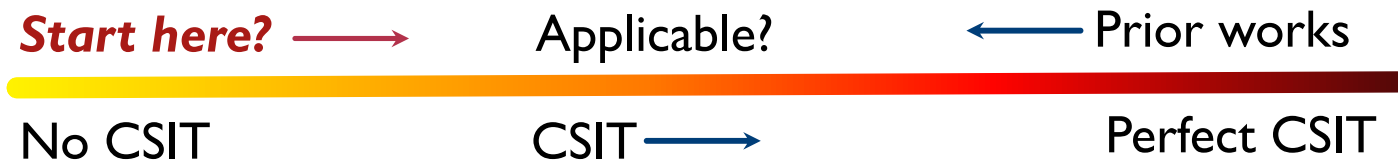
Interference alignment

- **Everyone gets half the cake** [Cadambe-Jafar'08]: $\text{DoF} = n/2$
 - **Diagonal** H_{ij} are time-varying and generic, $T \rightarrow \infty$, $m = T/2$ is almost surely asymptotically achievable
- **Remarks:**
 - Require very long block lengths
 - Require the channels to vary generically over time
 - Require full knowledge of the channel coefficients of *every link* in the network, at *each transmitter* and for *all times*!

Can we exploit the interference alignment principles in practical systems?

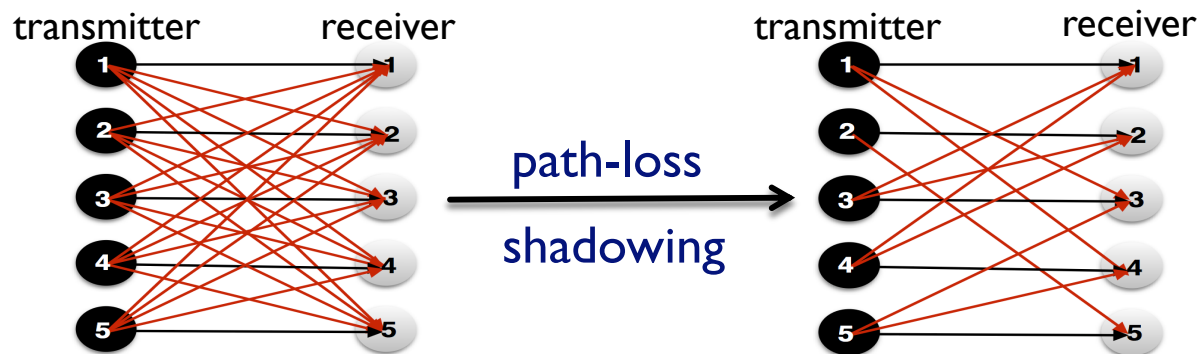
Interference management

- **Goal:** Exploit IA principles under realistic assumptions on CSIT
- **Prior works:** Abundant CSIT → relaxed CSIT
 - Perfect CSIT [Cadambe and Jafar, TIT 08]
 - Delayed CSIT [Maddah-Ali and Tse, TIT 12]
 - Alternating CSIT [Tandon, et al., TIT 13], partial and imperfect CSIT [Shi, et al., TSP 14],...
- **Curses:** CSIT is rarely abundant (due to training & feedback overhead)



Topological interference alignment

- **Blessings:** partial connectivity in dense wireless networks



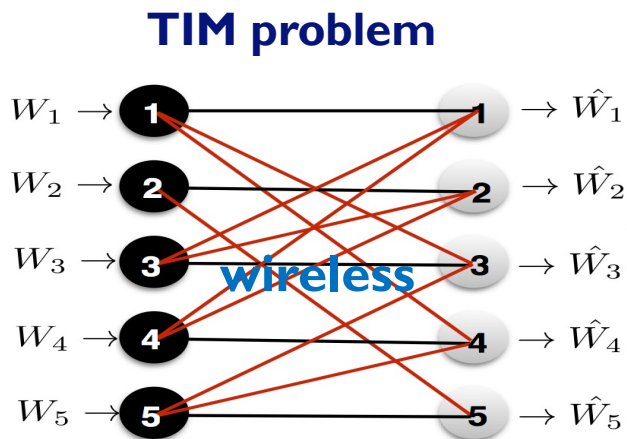
Degrees of Freedom?

$$\text{DoF} = \lim_{\text{SNR} \rightarrow \infty} \frac{C(\text{SNR})}{\log(\text{SNR})}$$

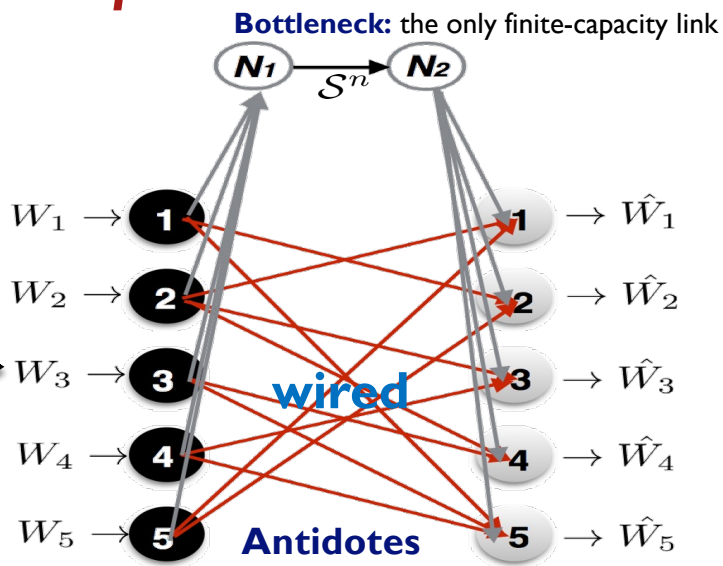
- **Approach:** topological interference management (TIM) [Jafar, TIT 14]
 - **Maximize the achievable DoF:** only based on the network topology information (**no CSIT**)

Index coding approach

- **Theorem** [Jafar, TIT 14]: under linear (vector space) solutions, TIM problem and index coding problem are **equivalent**



transformation
complements



Only a few index coding problems have been solved!

Low-rank matrix completion approach

- **Goal:** Deliver one data stream per user over N time slots

- Transmitter i transmits $\mathbf{v}_i s_i$, receiver i receives

$$\mathbf{y}_i = \mathbf{v}_i h_{ii} s_i + \sum_{(i,j) \in \mathcal{S}, i \neq j} \mathbf{v}_j h_{ij} s_j + \mathbf{n}_i \quad \mathcal{S}: \text{network connectivity pattern}$$

- Receiver decodes symbol s_i by projecting $\mathbf{y}_i$ into the space $\mathbf{u}_i \in \mathbb{C}^N$

$$\mathbf{u}_i^H \mathbf{y}_i = \mathbf{u}_i^H \mathbf{v}_i h_{ii} s_i + \sum_{(i,j) \in \mathcal{S}, i \neq j} \mathbf{u}_i^H \mathbf{v}_j h_{ij} s_j + \mathbf{u}_i^H \mathbf{n}_i$$

- **Topological interference alignment condition**

$$M_{ij} = \begin{cases} \mathbf{u}_i^H \mathbf{v}_i = 1, & \forall i, \\ \mathbf{u}_i^H \mathbf{v}_j = 0, & \forall i \neq j, (i, j) \in \mathcal{S}, \\ \star, & \text{otherwise.} \end{cases} \quad \longrightarrow \quad \begin{aligned} \mathbf{u}_i^H \mathbf{y}_i &= h_{ii} s_i + \mathbf{u}_i^H \mathbf{n}_i \\ \text{DoF} &= \frac{1}{\text{rank}(\mathbf{M})} = \frac{1}{N} \end{aligned}$$

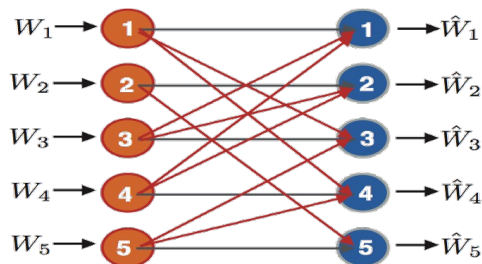
Generalized low-rank model

- Generalized low-rank optimization with network side information

$$\begin{aligned}
 & \underset{\mathbf{M} \in \mathbb{C}^{K \times K}}{\text{minimize}} \quad \text{rank}(\mathbf{M}) \\
 & \text{subject to} \quad \begin{aligned} & M_{ii} = 1, i = 1, \dots, K \\ & M_{ij} = 0, \forall (i, j) \in \mathcal{S} \end{aligned}
 \end{aligned}$$

topological interference alignment condition

- $\mathbf{M} = [\mathbf{u}_i^H \mathbf{v}_j]$: precoding vectors and decoding vectors $\mathbf{u}_k, \mathbf{v}_k \in \mathbb{C}^N$
- $\text{rank}(\mathbf{M})$ equals the inverse of achievable degrees-of-freedom (DoF) $\frac{1}{N}$



(a) Topological interference alignment

Transmitters

	1		0	0	
Receivers	1	1	0	0	
	0		1		0
	0			1	0
		0			1

side information $\mathcal{S}$

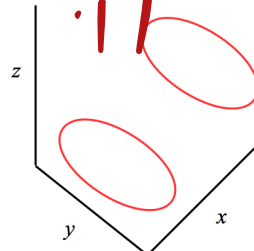
(b) Side information modeling matrix $\mathbf{M}$

Nuclear norm fails

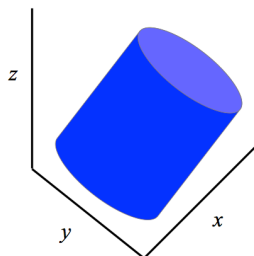
- **Convex relaxation fails:** always return the identity matrix!

$$\begin{aligned} & \underset{M \in \mathbb{C}^{K \times K}}{\text{minimize}} && \|M\|_* \\ & \text{subject to} && M_{ii} = 1, i = 1, \dots, K \\ & && M_{ij} = 0, \forall (i, j) \in \mathcal{S} \end{aligned}$$

$\Rightarrow M^* = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$



$$\begin{pmatrix} x & y \\ y & z \end{pmatrix}$$



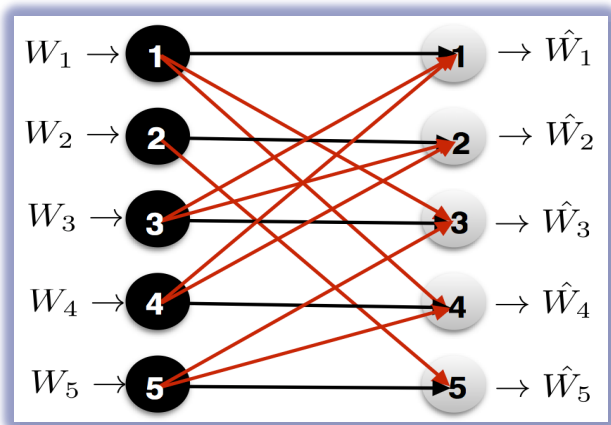
- **Fact:** $\text{Trace}(M) \leq \|M\|_*$
- **Proposal:** Solve the nonconvex problems directly with rank adaptivity

$$\begin{aligned} & \underset{M \in \mathbb{C}^{K \times K}}{\text{minimize}} && f(M) := \|\mathcal{A}(M) - z\|_F^2 \\ & \text{subject to} && \text{rank}(M) = r \end{aligned}$$

manifold constraint

Riemannian manifold
optimization problem

Numerical results



1		0	0	
	1	0	0	
0		1		0
0			1	0
	0			1

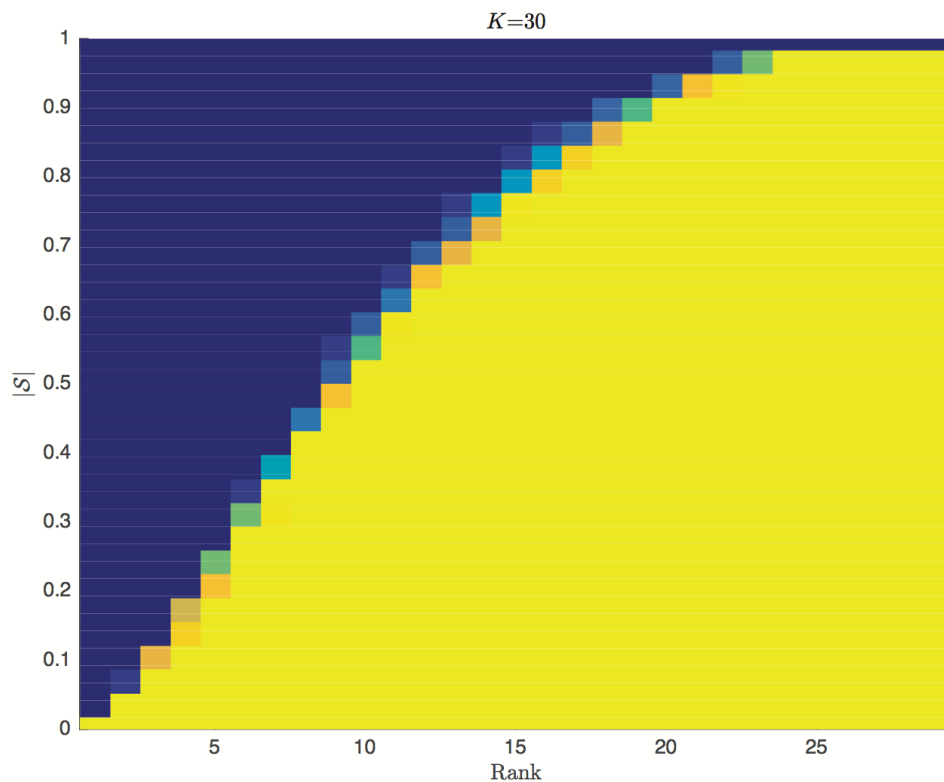
Recover all the optimal DoF results for the special TIM problems in [Jafar '14]

1	.1	0	0	9.5
6.8	1	0	0	64
0	.1	1	-1	0
0	-.1	-1	1	0
.1	0	-.1	.1	1



Provide numerical insights (optimal/lower-bound) for the general TIM problems

Phase transitions for topological IA



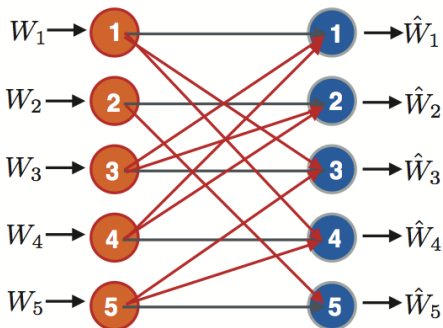
The heat map indicates the empirical probability of success (blue=0%; yellow=100%)

Summary: generalized low-rank models

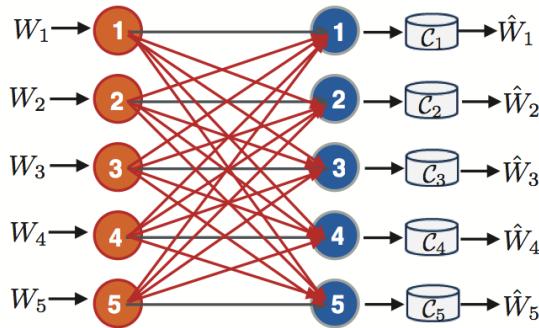
- Generalized low-rank optimization for dense edge networks

$$\underset{M \in \mathbb{C}^{m \times n}}{\text{minimize}} \quad \text{rank}(M) \quad \text{subject to} \quad M \in \mathcal{D}$$

- $\mathcal{D}$ encodes network side information, e.g., cached files, network topology, computed intermediate values for data shuffling



(a) TIM problem.



(b) Cache-aided interference channel.

Transmitters

Receivers

1		0	0	
	1	0	0	
0		1		0
0			1	0
	0			1

(c) Side information modeling matrix.

Concluding remarks

- **Structured sparse models**

- Group sparse optimization offers a principled way for network adaptation, e.g., to minimize network power consumption
- Sparsity control and estimation is powerful to support massive device connectivity

- **Future directions:**

- More application scenarios: IoTs, V2X ...

Concluding remarks

- **Generalized low-rank models**

- Low-rank matrix completion provides a systematic approach to investigate the topological interference alignment problem
- Low-rank model is powerful for performance optimization in mobile edge caching and distributed computing systems

- **Future directions:**

- More applications: blind deconvolution for IoT, big data analytics (e.g., ranking)

To learn more...

- **Web:** <http://shiyuanming.github.io/sparserank.html>
- **Papers:**
- Y. Shi, J. Zhang, and K. B. Letaief, “Group sparse beamforming for green Cloud-RAN,” *IEEE Trans. Wireless Commun.*, vol. 13, no. 5, pp. 2809-2823, May 2014. (The 2016 Marconi Prize Paper Award)
- Y. Shi, J. Zhang, B. O’Donoghue, and K. B. Letaief, “Large-scale convex optimization for dense wireless cooperative networks,” *IEEE Trans. Signal Process.*, vol. 63, no. 18, pp. 4729-4743, Sept. 2015. t. 2015. (The 2016 IEEE Signal Processing Society Young Author Best Paper Award)
- Y. Shi, J. Zhang, K. B. Letaief, B. Bai and W. Chen, “Large-scale convex optimization for ultra-dense Cloud-RAN,” *IEEE Wireless Commun. Mag.*, pp. 84-91, Jun. 2015.
- Y. Shi, J. Zhang, W. Chen, and K. B. Letaief, “Generalized sparse and low-rank optimization for ultra-dense networks,” *IEEE Commun. Mag.*, vol. 56, no. 6, pp. 42-48, Jun., 2018.

To learn more...

- Y. Shi, J. Zhang, and K. B. Letaief, “Optimal stochastic coordinated beamforming for wireless cooperative networks with CSI uncertainty,” *IEEE Trans. Signal Process.*, vol. 63,, no. 4, pp. 960-973, Feb. 2015.
- Y. Shi, J. Zhang, and K. B. Letaief, “Robust group sparse beamforming for multicast green Cloud- RAN with imperfect CSI,” *IEEE Trans. Signal Process.*, vol. 63, no. 17, pp. 4647-4659, Sept. 2015.
- Y. Shi, J. Cheng, J. Zhang, B. Bai, W. Chen and K. B. Letaief, “Smoothed L_p -minimization for green Cloud-RAN with user admission control,” *IEEE J. Select. Areas Commun.*, vol. 34, no. 4, Apr. 2016.
- Y. Shi, J. Zhang, and K. B. Letaief, “Low-rank matrix completion for topological interference management by Riemannian pursuit,” *IEEE Trans. Wireless Commun.*, vol. 15, no. 7, Jul. 2016.
- Y. Shi, B. Mishra, and W. Chen, “Topological interference management with user admission control via Riemannian optimization,” *IEEE Trans. Wireless Commun.*, vol. 16, no. 11, pp. 7362-7375, Nov. 2017.
- X. Peng, Y. Shi, J. Zhang, and K. B. Letaief, “Layered group sparse beamforming for cache-enabled wireless networks,” *IEEE Trans. Commun.*, vol. 65, no. 12, pp. 5589-5603, Nov. 2017.
- Y. Shi, J. Zhang, W. Chen, and K. B. Letaief, “Enhanced group sparse beamforming for dense green Cloud-RAN: A random matrix approach,” *IEEE Trans. Wireless Commun.*, vol. 17, no. 4, pp. 2511-2524, Apr. 2018.

Generalized Sparse and Low-Rank Optimization for Ultra-Dense Networks

Part II: Algorithms and Theory

Yuanming Shi



上海科技大学
ShanghaiTech University

Outline

- **Motivations**

- Issues on computation, storage, nonconvexity,...

- **Two Vignettes:**

- Large-scale convex optimization

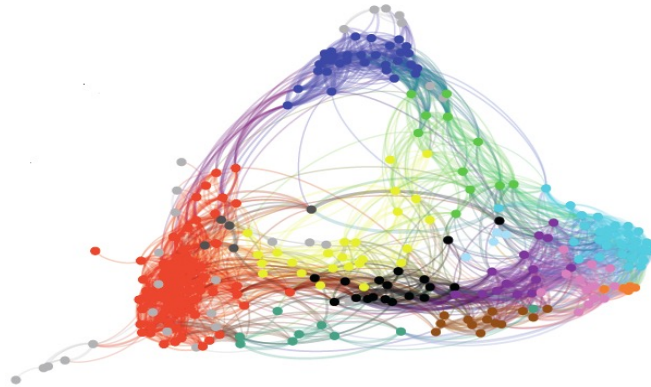
- Motivation: Why convex optimization?
 - Large-Scale Convex Optimization Algorithms

- Scalable nonconvex optimization on manifolds

- Motivation: Why Nonconvex Optimization?
 - Riemannian Optimization Algorithms

- **Future Directions**

Motivation: **Optimization** for **Dense Wireless Networks**

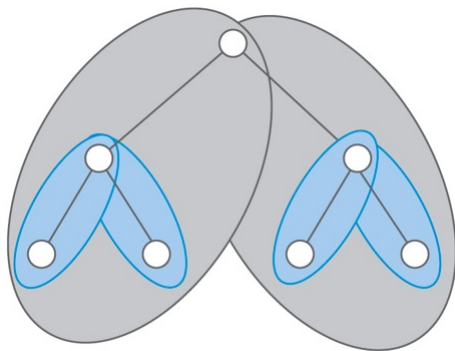


Motivations

- **The era of dense wireless networks**
 - Lead to new issues related to modeling and computing
- **Part I: Modeling issue**
 - Sparse and low-rank modeling frameworks for dense wireless networks
- **Part II: Computational issue**
 - Excessively large problem dimension, parameter size
 - Real-time communication requirements: polynomial-time algorithms often **not fast enough**
 - Non-convexity in general formulations

Issue A: Large-scale structured optimization

- Explosion in scale and complexity of the optimization problem in dense wireless networks



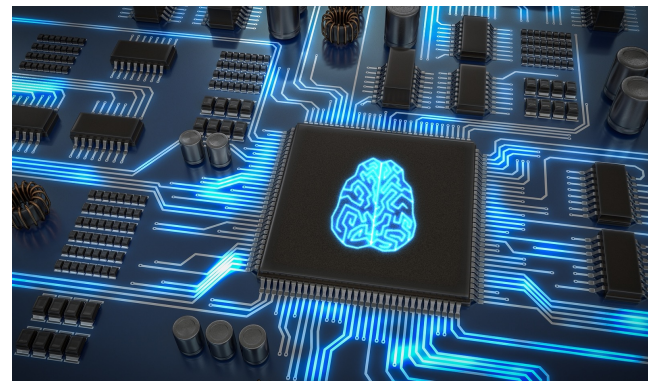
1		0	0	
	1	0	0	
0		1		0
0			1	0
	0			1

- **Questions:**

- How to exploit the low-dimensional structures (e.g., sparsity and low-rankness) to assist efficient algorithms design?

Issue B: Real-time convex optimization

- Polynomial-time algorithms often not fast enough for real-time communications: **parallel computing and approximations are essential**



- **Questions:**

- When is there a gap between polynomial-time and exponential-time algorithms?
- How to reduce computational complexity while retaining optimality and accuracy?

Issue C: Scalable nonconvex optimization

- Nonconvex optimization may be super scary

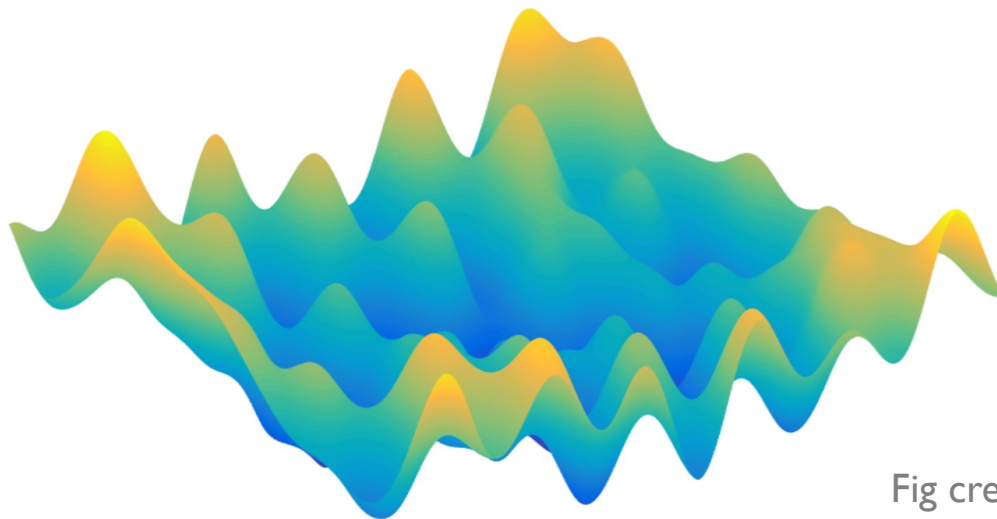


Fig credit: Chen

- **Question:**
 - How to exploit the geometry of nonconvex programs to guarantee optimality and enable scalability in computation and storage?

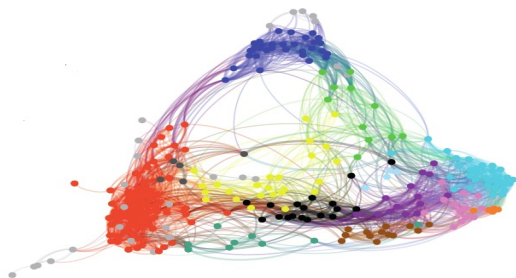
Vignettes A: **Large-Scale Convex** Optimization

1. Motivation: Why Convex Optimization?

- 1) Theory I: Convexify sparse functions
- 2) Theory II: Geometry of convex relaxation

2. Large-Scale Convex Optimization Algorithms

- 1) Matrix stuffing for homogeneous self-dual embedding transforming
- 2) Operator splitting for homogeneous self-dual embedding solving



*Motivation: Why **Convex** **Optimization**?*

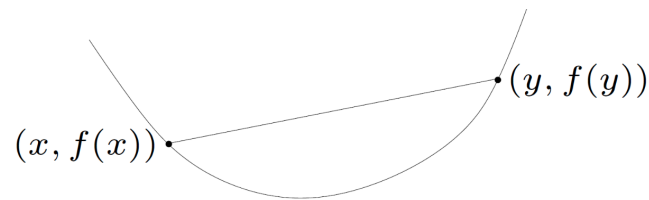
Convex optimization – classical form

- Convex optimization problem in classical form

$$\begin{aligned} & \underset{\mathbf{z}}{\text{minimize}} && f_0(\mathbf{z}; \boldsymbol{\alpha}) \\ & \text{subject to} && f_i(\mathbf{z}; \boldsymbol{\alpha}) \leq g_i(\mathbf{z}; \boldsymbol{\alpha}), i = 1, \dots, m \\ & && u_i(\mathbf{z}; \boldsymbol{\alpha}) = v_i(\mathbf{z}; \boldsymbol{\alpha}), i = 1, \dots, p. \end{aligned}$$

- f_i convex, g_i concave, u_i, v_i affine
- **Convex functions:** have nonnegative (upward) curvature

$$f_i(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f_i(\mathbf{x}) + (1 - \theta) f_i(\mathbf{y})$$



Convex optimization – conic form

- Convex optimization in *modern* canonical form

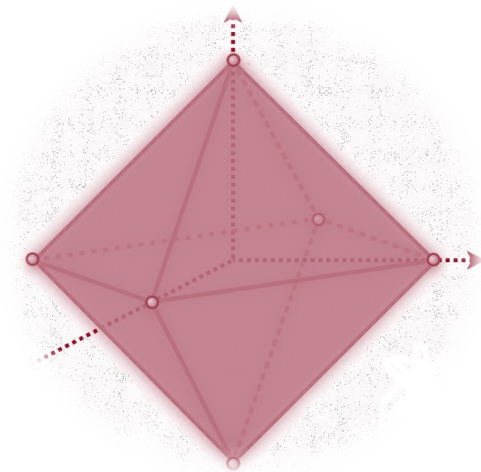
$$\begin{aligned} & \underset{\boldsymbol{\nu}, \boldsymbol{\mu}}{\text{minimize}} && \mathbf{c}^T \boldsymbol{\nu} \\ & \text{subject to} && \mathbf{A}\boldsymbol{\nu} + \boldsymbol{\mu} = \mathbf{b} \\ & && (\boldsymbol{\nu}, \boldsymbol{\mu}) \in \mathbb{R}^n \times \mathcal{K}. \end{aligned}$$

- $\mathcal{K} = \mathcal{K}_1 \times \cdots \times \mathcal{K}_q \in \mathbb{R}^m$ is a Cartesian product of closed convex cones
 - **Nonnegative reals:** $\mathbb{R}_+ = \{z \in \mathbb{R} | z \geq 0\}$ (LP)
 - **Second-order cone:** $\mathcal{Q}^d = \{(z, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^{d-1} | \|\mathbf{x}\| \leq z\}$ (SOCP)
 - **Positive semidefinite cone:** $\mathbf{S}_+^n = \{\mathbf{M} \in \mathbb{R}^{n \times n} | \mathbf{M} = \mathbf{M}^T, \mathbf{M} \succeq \mathbf{0}\}$ (SDP)

Why?

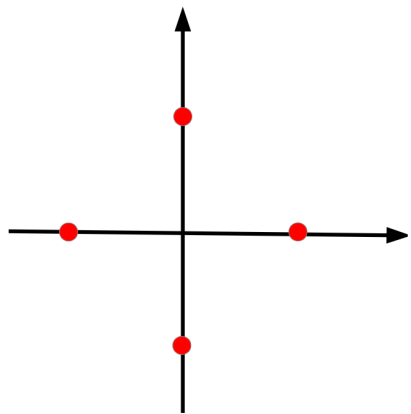
- **Theoretical foundations:** Beautiful, nearly complete theory
 - Duality, optimality conditions, convex geometry,...
- **Effective algorithms:** Convex optimization problems can be solved effectively with global optimality
 - Use generic methods for not huge problems: high level language support (CVX/CVXPY/Convex.jl) makes prototyping easy
 - Develop custom methods for huge problems (e.g., stochastic gradient descent)
- **Lots of applications:** Machine learning, signal processing, statistics, wireless communications, ...

Theory I: **Convexify** **Sparse** Functions

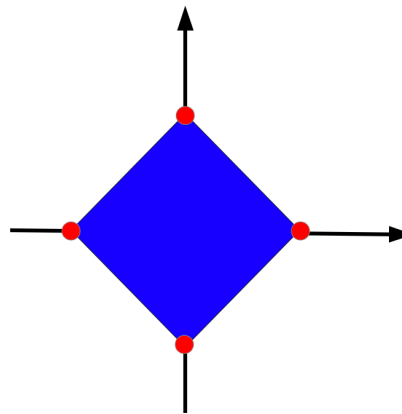


Geometric view: sparsity

- Sparse approximation via convex hull $\mathcal{D} := \text{conv}(\{\pm \mathbf{e}_i | i \in [n]\})$



1-sparse vectors of
Euclidean norm 1

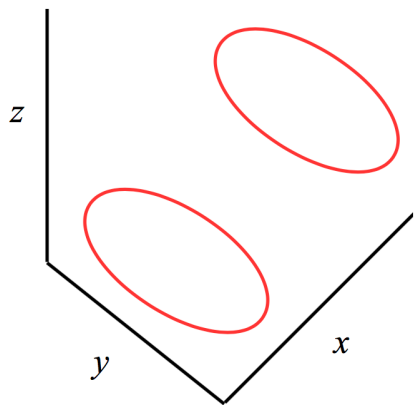


convex hull: ℓ_1 -norm

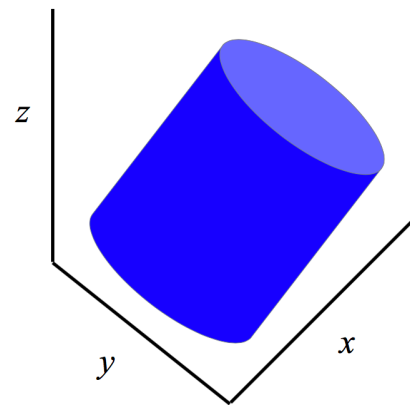
$$\|z\|_1 = \sum_{i=1}^n |z_i|$$

Geometric view: low-rank

- Low-rank approximation via convex hull



$$\begin{pmatrix} x & y \\ y & z \end{pmatrix}$$



2x2 rank 1 symmetric
matrices (normalized)

convex hull: nuclear norm

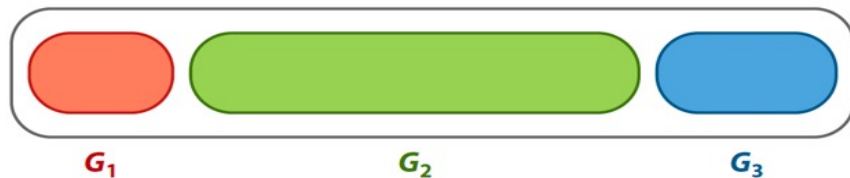
$$\|M\|_* = \sum_i \sigma_i(M)$$


Structured sparsity

- ℓ_p -regularized combinatorial penalties of the form

$$F_p(z) = \mu F(\text{Supp}(z)) + \nu \|z\|_p^p$$

- μ and ν are positive scalar coefficients, $p \in (1, \infty]$
- Positive-valued set-function F : control the structure of a model with non-zero patterns
- ℓ_p -norm: control the magnitude of the coefficients
- **Examples:** 1) individual sparsity $F(A) = |A|$; 2) group sparsity



$$F(A) = \sum_{i=1}^T 1_{\{A \cap G_j \neq \emptyset\}}$$

Structure preserved by convex relaxations

- The tightest positively homogeneous lower bound ($1/p + 1/q = 1$)

$$F_h(z) = (q\mu)^{1/q} (p\nu)^{1/p} Q(z)$$

- The convex envelope of Q is given by the norm Ω_p with dual norm as

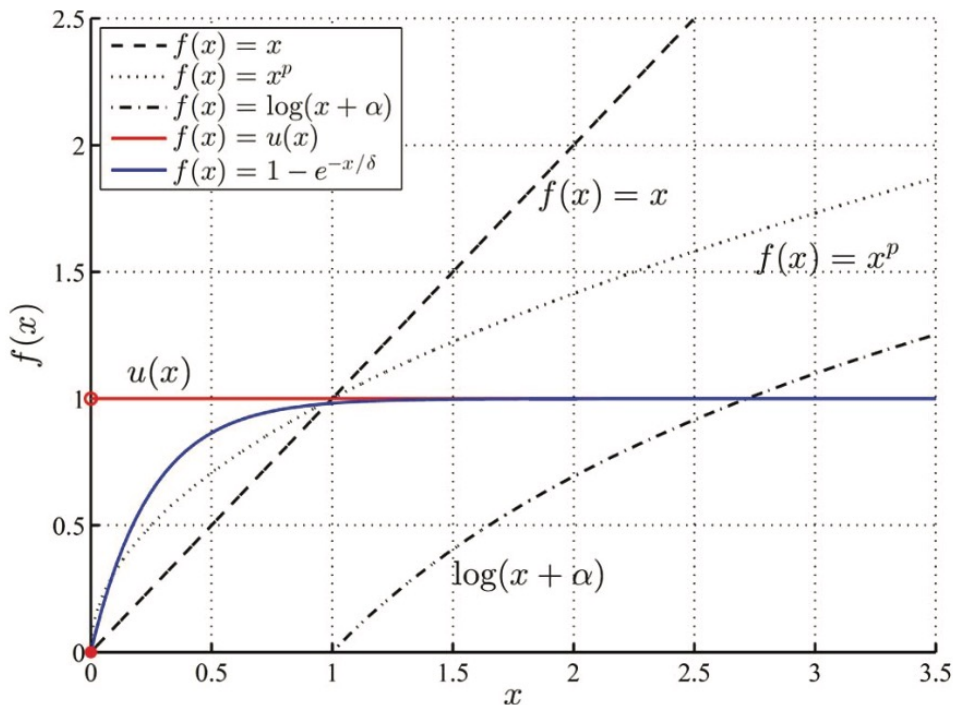
$$\Omega_p^*(s) := \max_{A \subset V, A \neq \emptyset} \frac{\|s_A\|_q}{F(A)^{1/q}}$$

- **Examples:**

- 1) ℓ_1 -norm (Lasso): If $F(A) = |A|$, then $\Omega_p(z) = \|z\|_1$, since $\Omega_p^*(s) = \|s\|_\infty$
- 2) ℓ_p -norm: If $F(A) = 1_{\{A \neq \emptyset\}}$, then $\Omega_p(z) = \|z\|_p$, since $\Omega_p^*(s) = \|s\|_q$
- 3) ℓ_1/ℓ_p -norm (Group Lasso): If $F(A) = \sum_{i=1}^T 1_{\{A \cap G_i \neq \emptyset\}}$, then $\Omega_p(z) = \sum_{i=1}^T \|z_{G_i}\|_p$

Enhance sparsity via sequential convex programming

- Goal: Provide tight approximation for sparsity function $u(x) = 1_{\{x \neq 0\}}$



Non-convex approximation:

$$\|\mathbf{x}\|_0 = \lim_{p \rightarrow 0} \|\mathbf{x}\|_p^p = \lim_{p \rightarrow 0} \sum |x_i|^p$$

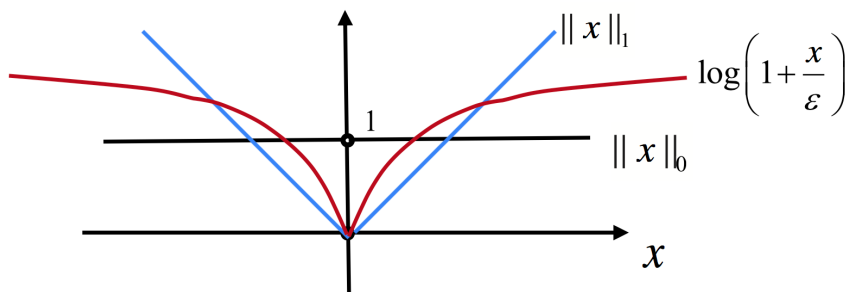
At the origin, ℓ_0 function is better approximated by the log-sum function
(check the slop at the origin)

Iterative reweighted- ℓ_1 algorithm (I)

- Consider the following (non-convex) sparse optimization problem

$$\underset{z \in \mathbb{C}^n}{\text{minimize}} \quad \|z\|_0 \quad \text{subject to} \quad z \in \mathcal{C}, z \succeq 0$$

- Approximate $\text{card}(z) \approx \log(1 + z/\epsilon)$, where $\epsilon > 0, z \in \mathbb{R}_+$



- Using this approximation, we get (non-convex) problem

$$\underset{z \in \mathbb{C}^n}{\text{minimize}} \quad \sum_{i=1}^n \log(1 + z_i/\epsilon) \quad \text{subject to} \quad z \in \mathcal{C}, z \succeq 0$$

Iterative reweighted- ℓ_1 algorithm (II)

- Find a local solution by linearizing objective at current point

$$\sum_{i=1}^n \log(1 + z_i/\epsilon) \approx \sum_{i=1}^n \log(1 + z_i^{[k]}/\epsilon) + \sum_{i=1}^n \frac{z_i - z_i^{[k]}}{\epsilon + z_i^{[k]}}$$

- **Solve resulting convex problem**

$$\underset{\mathbf{z} \in \mathbb{C}^n}{\text{minimize}} \quad \sum_{i=1}^n \omega_i^{[k]} z_i \quad \text{subject to} \quad \mathbf{z} \in \mathcal{C}, \mathbf{z} \succeq \mathbf{0}$$

with $\omega_i^{[k]} = 1/(\epsilon + x_i^{[k]})$, to get next iterate

- Repeat until convergence to get a local solution

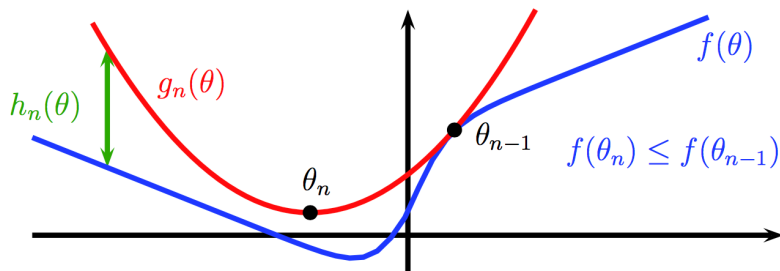
Iterative reweighted- ℓ_2 algorithm

- Adopt $\|z\|_p$ ($0 < p < 1$) to approximate $\|z\|_0$: $\|z\|_0 = \lim_{p \rightarrow 0} \|z\|_p^p$
- Solve the following (non-convex) smoothed ℓ_p -minimization problem

$$\underset{z \in \mathbb{C}^n}{\text{minimize}} \quad \sum_{i=1}^n (z_i^2 + \epsilon^2)^{p/2} \quad \text{subject to } z \in \mathcal{C}$$

- Construct an upper bound for objective function $Q(z; \omega^{[k]}) := \sum_{i=1}^n \omega_i^{[k]} z_i^2$

$$\omega_i^{[k]} = \frac{p}{2} \left[\left(z_i^{[k]} \right)^2 + \epsilon^2 \right]^{\frac{p}{2}-1}$$

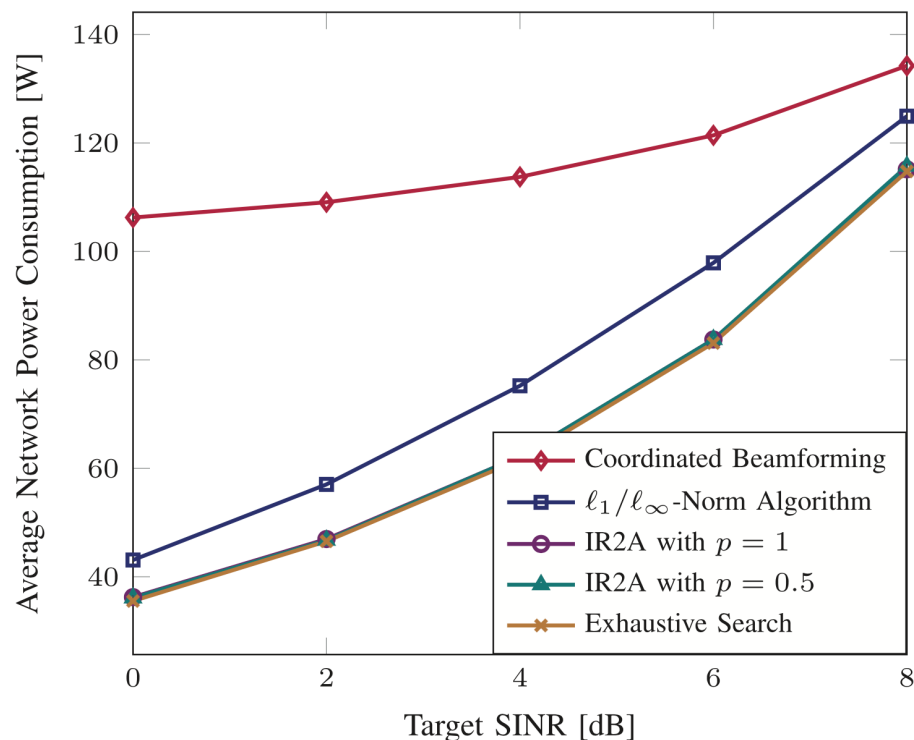


majorization-minimization algorithm

- Find the local solution via **convex** iterates $z^{[k+1]} := \arg \min_{z \in \mathcal{C}} Q(z; \omega^{[k]})$

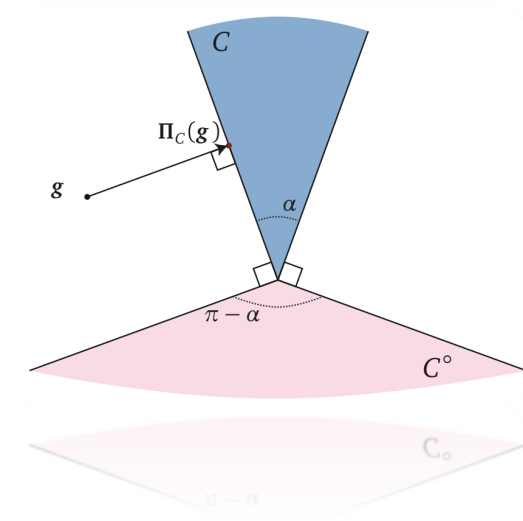
Simulation results: enhanced sparsity

- Network power minimization via group sparse beamforming



Group sparse beamforming
for network power
minimization (IR2A: iterative
reweighted ℓ_2 -algorithm)

Theory II: **Geometry** of **Convex Relaxation**



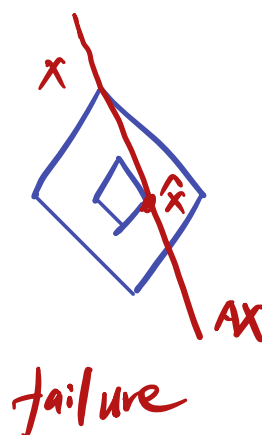
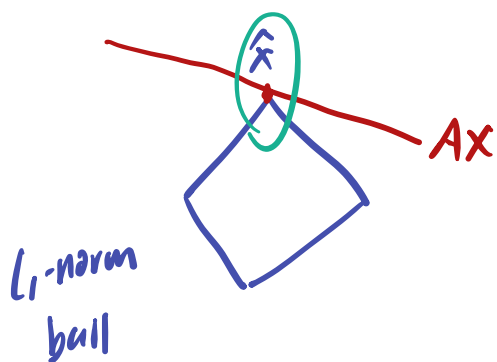
Linear inverse problems

- Let $x^\natural \in \mathbb{R}^d$ be a structured, unknown vector
 - Group sparsity for user activity detection
- Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex function that reflects structure, e.g. ℓ_1 -norm
- Let $A \in \mathbb{R}^{m \times d}$ be a measurement operator
- **Observe** $z = Ax^\natural$
- Find estimate $\hat{x}$ by solving **convex program**
$$\underset{x}{\text{minimize}} \quad \underbrace{f(x)} \quad \text{subject to} \quad \underbrace{Ax = z}$$
- **Hope:** $\hat{x} = x^\natural$

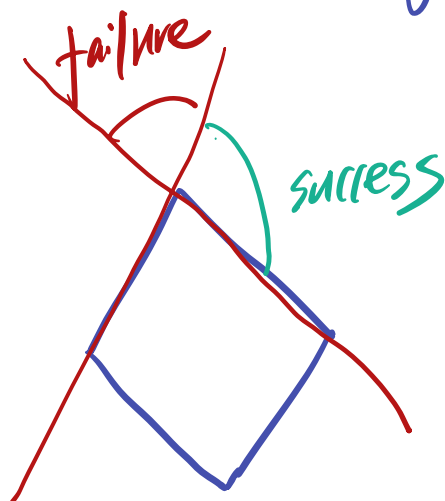
$$\underset{x \in \mathbb{R}^p}{\text{minimize}} \quad \|x\|_1$$

$$\text{subject to} \quad y = Ax \in \mathbb{R}^n$$

what is the probability that ℓ_1 -minimization succeeds in recovering x ?



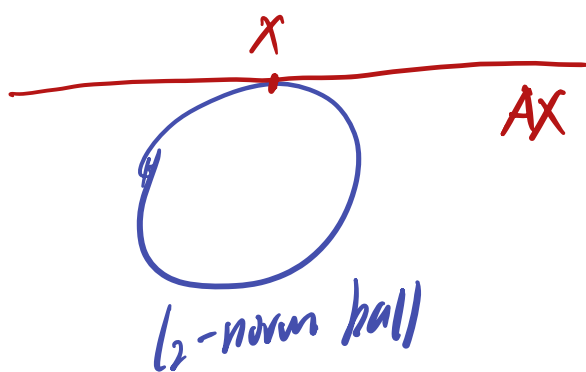
- suppose A is randomly generated



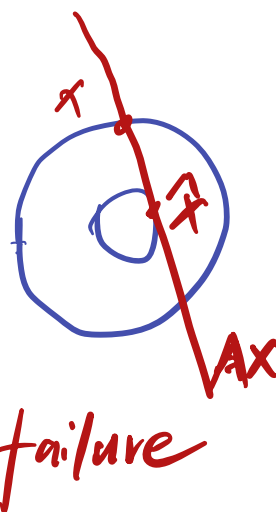
- if $x = [0, 1]^T$, and $A = [a_1, a_2] \in \mathbb{R}^{1 \times 2}$ (i.i.d. Gaussian),
 $P\{\ell_1\text{-min succeeds}\} = \frac{1}{2}$

Suppose A is randomly generated (i.i.d. Gaussian)
and consider

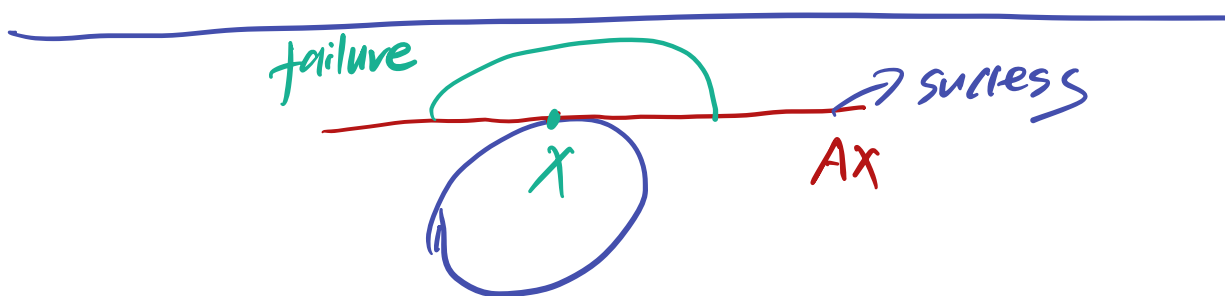
$$\begin{array}{ll} \text{minimize} & \|x\|_2 \\ \text{subject to} & y = Ax \in \mathcal{P}^n \end{array}$$



success



failure



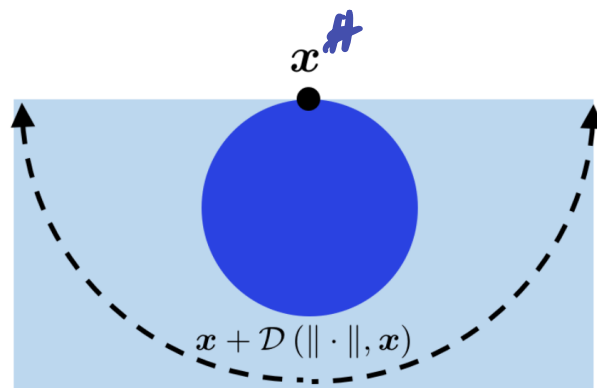
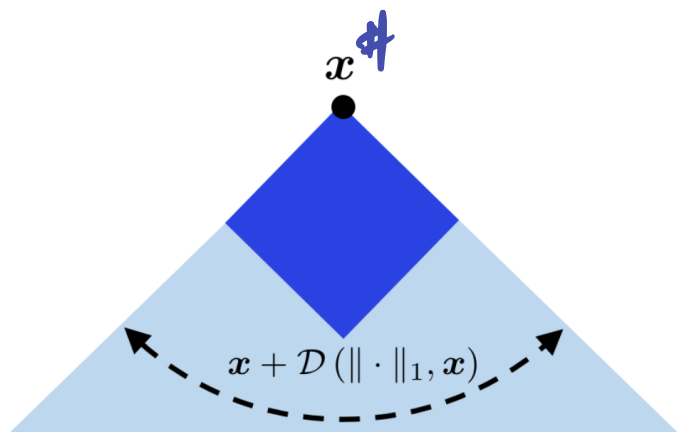
- if $x = [0, 1]^T$, and $A = [a_1, a_2] \in \mathbb{R}^{1 \times 2}$ (i.i.d. Gaussian),

$$P\{\text{\textcolor{red}{l}_2\text{-min succeeds}\} = 0$$

Geometry of linear inverse problems

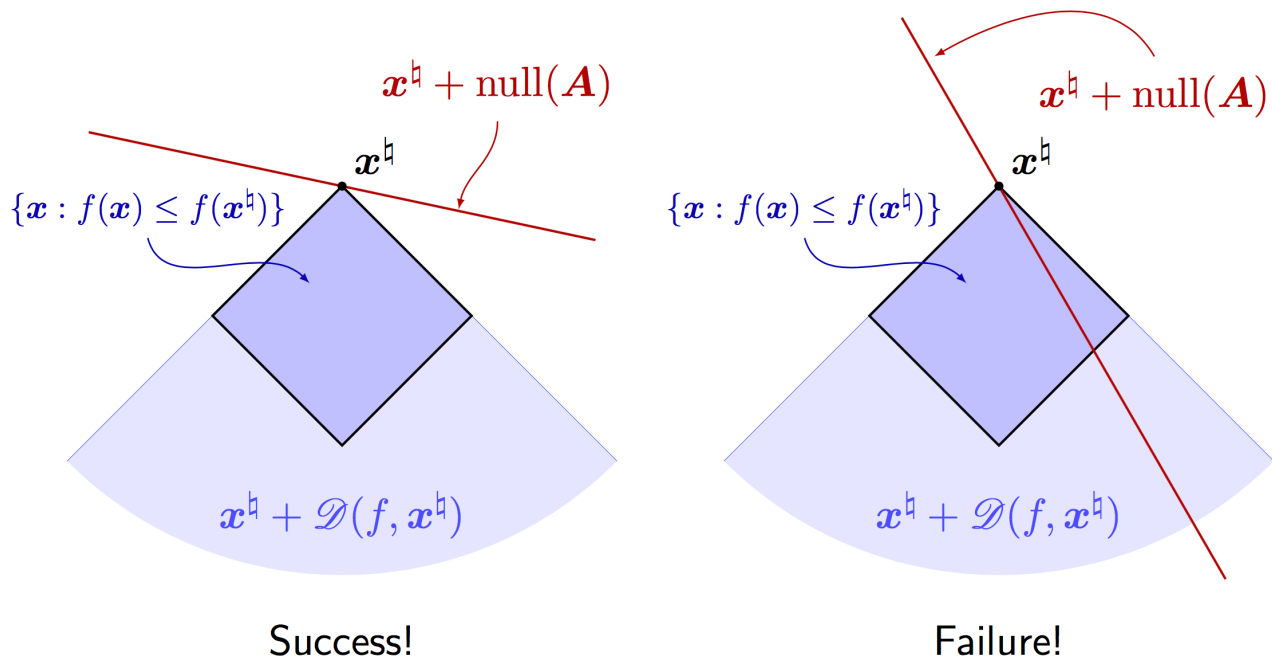
- **Descent cone** of a function f at a point z is

$$\mathcal{D}(f, z) := \{d : f(z + \epsilon d) \leq f(z), \text{ for some } \epsilon > 0\}$$



References: Rockafellar 1970

Geometry of linear inverse problems



References: Candes–Romberg–Tao 2005, Rudelson–Vershynin 2006, Chandrasekaran et al. 2010, Amelunxen et al. 2013

Linear inverse problems with random data

■ Assume

- The vector $\mathbf{x}^\natural \in \mathbb{R}^d$ is unknown
- The observation $\mathbf{z} = \mathbf{A}\mathbf{x}^\natural$ where $\mathbf{A} \in \mathbb{R}^{m \times d}$ is standard normal
- The vector $\hat{\mathbf{x}}$ solves

$$\text{minimize } f(\mathbf{x}) \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{z}$$

■ Then

$$m \gtrsim \delta(\mathcal{D}(f, \mathbf{x}^\natural)) \implies \hat{\mathbf{x}} = \mathbf{x}^\natural, \quad \text{w.h.p.}$$

$$m \lesssim \delta(\mathcal{D}(f, \mathbf{x}^\natural)) \implies \hat{\mathbf{x}} \neq \mathbf{x}^\natural, \quad \text{w.h.p.}$$


statistical dimension

Examples for statistical dimension

- **Example 1:** ℓ_1 -minimization for compressed sensing

- $\mathbf{x}^\natural \in \mathbb{R}^d$ with s non-zero entries

$$\delta \left(\mathcal{D}(\|\cdot\|_1, \mathbf{x}^\natural) \right) = \inf_{\tau \geq 0} \left\{ s(1 + \tau^2) + (d - s) \sqrt{\frac{2}{\pi}} \int_{\tau}^{\infty} (z - \tau)^2 e^{-z^2} dz \right\}$$

- **Example 2:** ℓ_1/ℓ_2 -minimization for massive device connectivity

- $\mathbf{X}^\natural \in \mathbb{R}^{N \times M}$ with s non-zero rows

$$\delta \left(\mathcal{D}(\|\cdot\|_{2,1}, \mathbf{X}^\natural) \right) = \inf_{\tau \geq 0} \left\{ s(M + \tau^2) + (N - s) \frac{2^{1-M/2}}{\Gamma(M/2)} \int_{\tau}^{\infty} (u - \tau)^2 u^{M-1} e^{-\frac{u^2}{2}} du \right\}$$

Numerical phase transition

- Compressed sensing with ℓ_1 -minimization

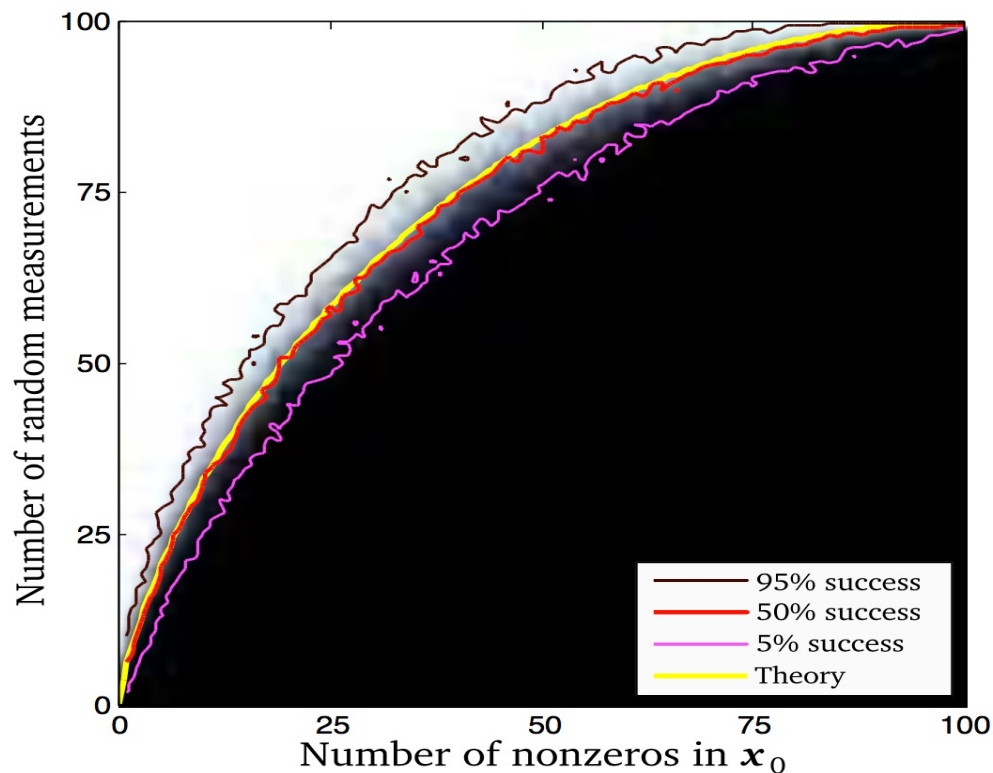
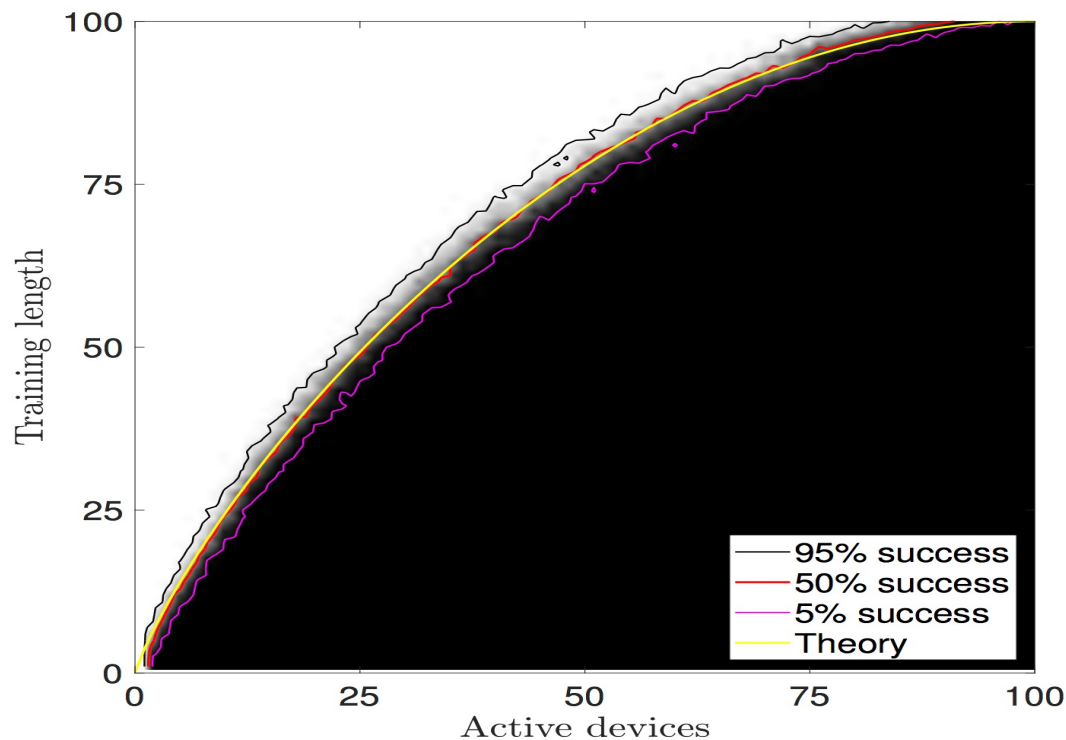


Figure credit: Amelunxen-McCoy-Tropp'13

Numerical phase transition

- User activity detection via ℓ_1/ℓ_2 -minimization



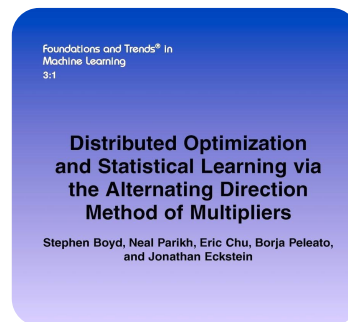
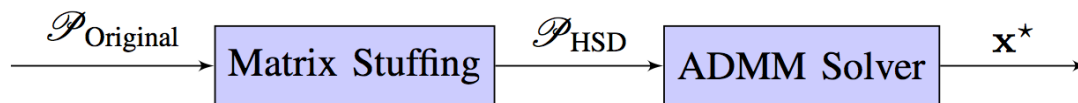
group-structured
sparsity estimation

Summary of convex optimization

- Theoretical foundations for sparse optimization
 - Convex relaxation: convex hull, convex analysis
 - Fundamental bounds for convex methods: convex geometry, high-dimensional statistics
- Computational limits for (convexified) sparse optimization
 - Custom methods (e.g., stochastic gradient descent): not generalizable for complicated problems
 - Generic methods (e.g., CVX): not scalable to large problem sizes

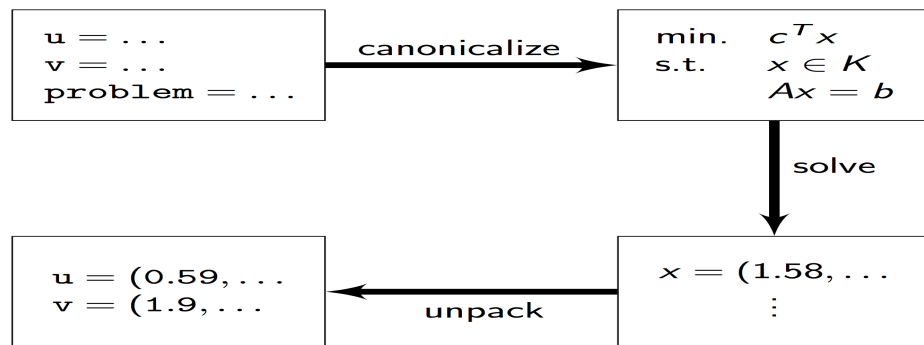
Can we design a unified framework for general large-scale convex programs?

Large-Scale *Convex Optimization* Algorithms



Modeling languages

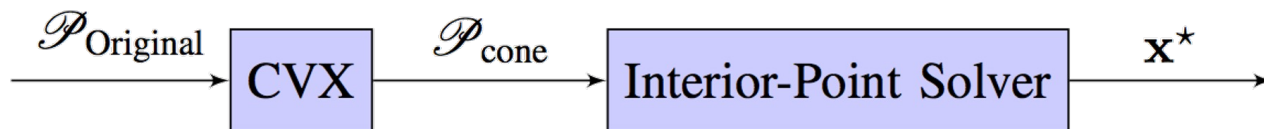
- High level language support for convex optimization
 - **Stage one:** problem description automatically transformed to standard form
 - **Stage two:** solved by standard solver, transformed back to original form



- **Implementation:** YALMIP, CVX (Matlab), CVXPY (Python), Convex.jl (Julia)

Modeling languages

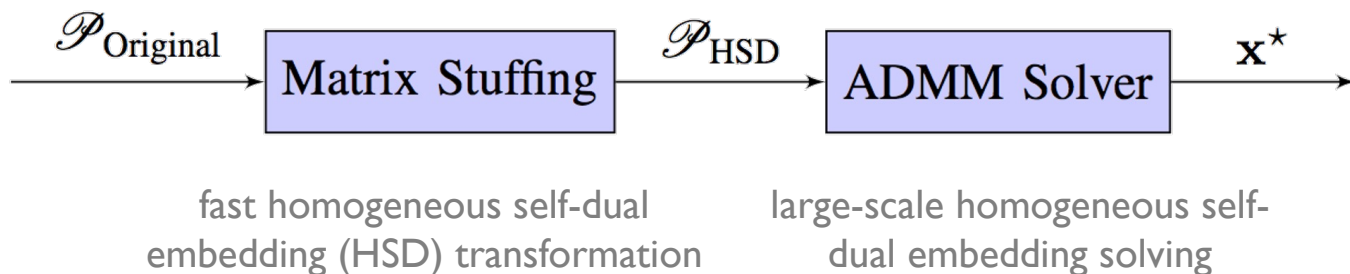
- Disciplined convex programming framework [Grant & Boyd '08]



- enable rapid prototyping (for small and medium problems)
- widely used for applications with medium scale problems
- shifts focus from *how to solve* to *what to solve*
- **Large-scale problems:** time consuming in modeling phase & solving phase
- **Goal:** Scale to large problem sizes in modeling phase and solving phase

Large-scale convex optimization

- **Proposal:** Two-stage approach for large-scale convex optimization



- **Matrix stuffing:** Fast homogeneous self-dual embedding (HSD) transformation
- **Operator splitting (ADMM):** Large-scale homogeneous self-dual embedding

*Stage I: **Matrix** **Stuffing***

Smith form reformulation

- **Goal:** transform the classical form to conic form

$$\begin{array}{ll} \underset{\mathbf{z}}{\text{minimize}} & f_0(\mathbf{z}; \boldsymbol{\alpha}) \\ \text{subject to} & f_i(\mathbf{z}; \boldsymbol{\alpha}) \leq g_i(\mathbf{z}; \boldsymbol{\alpha}), \\ & u_i(\mathbf{z}; \boldsymbol{\alpha}) = v_i(\mathbf{z}; \boldsymbol{\alpha}). \end{array} \quad \Rightarrow \quad \begin{array}{ll} \underset{\boldsymbol{\nu}, \boldsymbol{\mu}}{\text{minimize}} & \mathbf{c}^T \boldsymbol{\nu} \\ \text{subject to} & \mathbf{A}\boldsymbol{\nu} + \boldsymbol{\mu} = \mathbf{b}, \\ & (\boldsymbol{\nu}, \boldsymbol{\mu}) \in \mathbb{R}^n \times \mathcal{K}. \end{array}$$

- **Key idea:** Introduce a new variable for each subexpression in classical form [Smith '96]
 - The Smith form is ready for standard cone programming transformation

Example

- Coordinated beamforming problem **family**

$$\mathcal{P}_{\text{Original}} : \text{minimize } \|\mathbf{v}\|_2^2$$

$$\text{subject to } \|\mathbf{D}_l \mathbf{v}\|_2 \leq \sqrt{P_l}, \forall l, \quad \text{Per-BS power constraint} \quad (1)$$

$$\|\mathbf{C}_k \mathbf{v} + \mathbf{g}_k\|_2 \leq \beta_k \mathbf{r}_k^T \mathbf{v}, \forall k. \quad \text{QoS constraints} \quad (2)$$

- Smith form reformulation

$$\mathcal{G}_1(l) : \begin{cases} (y_0^l, \mathbf{y}_1^l) \in \mathcal{Q}^{KN_l+1} \\ y_0^l = \sqrt{P_l} \in \mathbb{R} \\ \mathbf{y}_1^l = \mathbf{D}_l \mathbf{v} \in \mathbb{R}^{KN_l} \end{cases}$$

Smith form for (1)

$$\mathcal{G}_2(k) : \begin{cases} (t_0^k, \mathbf{t}_1^k) \in \mathcal{Q}^{K+1} \\ t_0^k = \beta_k \mathbf{r}_k^T \mathbf{v} \in \mathbb{R} \\ \mathbf{t}_1^k = \mathbf{t}_2^k + \mathbf{t}_3^k \in \mathbb{R}^{K+1} \\ \mathbf{t}_2^k = \mathbf{C}_k \mathbf{v} \in \mathbb{R}^{K+1} \\ \mathbf{t}_3^k = \mathbf{g}_k \in \mathbb{R}^{K+1} \end{cases}$$

Smith form for (2)

The Smith form is readily to be reformulated as the standard cone program

Optimality condition

- KKT conditions (necessary and sufficient, assuming strong duality)
 - Primal feasibility: $\mathbf{A}\boldsymbol{\nu}^* + \boldsymbol{\mu}^* - \mathbf{b} = \mathbf{0}$
 - Dual feasibility: $\mathbf{A}^T \boldsymbol{\eta}^* - \boldsymbol{\lambda}^* + \mathbf{c} = \mathbf{0}$
 - Complementary slackness: $\mathbf{c}^T \boldsymbol{\nu}^* + \mathbf{b}^T \boldsymbol{\eta}^* = 0$ zero duality gap
 - Feasibility: $(\boldsymbol{\nu}^*, \boldsymbol{\mu}^*, \boldsymbol{\lambda}^*, \boldsymbol{\eta}^*) \in \mathbb{R}^n \times \mathcal{K} \times \{0\}^n \times \mathcal{K}^*$

no solution if primal or dual problem infeasible/unbounded

Homogeneous self-dual (HSD) embedding

- **HSD embedding** of the primal-dual pair of transformed standard cone program (based on KKT conditions)

$$\begin{array}{l} \text{minimize}_{\nu, \mu} \mathbf{c}^T \nu \\ \text{subject to } \mathbf{A}\nu + \mu = \mathbf{b} \\ (\nu, \mu) \in \mathbb{R}^n \times \mathcal{K}. \end{array} + \begin{array}{l} \text{maximize}_{\eta, \lambda} -\mathbf{b}^T \eta \\ \text{subject to } -\mathbf{A}^T \eta + \lambda = \mathbf{c} \\ (\lambda, \eta) \in \{0\}^n \times \mathcal{K}^* \end{array} \Rightarrow \begin{array}{l} \mathcal{F}_{\text{HSD}} : \text{find } (\mathbf{x}, \mathbf{y}) \\ \text{subject to } \mathbf{y} = \mathbf{Q}\mathbf{x} \\ \mathbf{x} \in \mathcal{C}, \mathbf{y} \in \mathcal{C}^* \end{array}$$

$$\underbrace{\begin{bmatrix} \lambda \\ \mu \\ \kappa \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{A}^T & \mathbf{c} \\ -\mathbf{A} & \mathbf{0} & \mathbf{b} \\ -\mathbf{c}^T & -\mathbf{b}^T & \mathbf{0} \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \nu \\ \eta \\ \tau \end{bmatrix}}_{\mathbf{x}} \quad \text{finding a } \textit{nonzero} \text{ solution}$$

- This feasibility problem is homogeneous and self-dual

Recovering solution or certificates

- Any HSD solution $(\nu, \mu, \lambda, \eta, \tau, \kappa)$ falls into one of three cases:
 - **Case 1:** $\tau > 0, \kappa = 0$, then $\hat{\nu} = \nu/\tau, \hat{\eta} = \eta/\tau, \hat{\mu} = \mu/\tau$ is a solution
 - **Case 2:** $\tau = 0, \kappa > 0$, implies $\mathbf{c}^T \nu + \mathbf{b}^T \eta < 0$
 - If $\mathbf{b}^T \eta < 0$, then $\hat{\eta} = \eta/(-\mathbf{b}^T \eta)$ certifies primal infeasibility
 - If $\mathbf{c}^T \nu < 0$, then $\hat{\nu} = \nu/(-\mathbf{c}^T \nu)$ certifies dual infeasibility
 - **Case 3:** $\tau = \kappa = 0$, nothing can be said about original problem
- **HSD embedding:** 1) obviates need for phase I / phase II solves to handle infeasibility/unboundedness; 2) used in all interior-point cone solvers

Matrix stuffing for fast transformation

- HSD embedding of the primal-dual pair of standard cone program

$$\underbrace{\begin{bmatrix} \lambda \\ \mu \\ \kappa \end{bmatrix}}_y = \underbrace{\begin{bmatrix} 0 & \mathbf{A}^T & \mathbf{c} \\ -\mathbf{A} & 0 & \mathbf{b} \\ -\mathbf{c}^T & -\mathbf{b}^T & 0 \end{bmatrix}}_Q \underbrace{\begin{bmatrix} \nu \\ \eta \\ \tau \end{bmatrix}}_x$$

- **Matrix stuffing:** fast HSD embedding transformation
 - Generate and keep the structure Q
 - Copy problem instance parameters to update the entries in Q

Stage II: **Operator Splitting**

$$\begin{aligned} \mathcal{F}_{\text{HSD}} : \text{find } & (\mathbf{x}, \mathbf{y}) \\ \text{subject to } & \mathbf{y} = \mathbf{Q}\mathbf{x} \\ & \mathbf{x} \in \mathcal{C}, \mathbf{y} \in \mathcal{C}^* \end{aligned}$$

Alternating direction method of multipliers

- **ADMM**: an operator splitting method solving convex problems in form

$$\mathcal{P}_{\text{ADMM}} : \text{minimize } f(\mathbf{x}) + g(\mathbf{z}) \quad \text{subject to } \mathbf{x} = \mathbf{z}$$

- f, g convex, **not necessarily smooth**, can take infinite values
- The basic ADMM algorithm is

$$\mathbf{x}^{[k+1]} = \arg \min_{\mathbf{x}} \left(f(\mathbf{x}) + (\rho/2) \|\mathbf{x} - \mathbf{z}^{[k]} - \lambda^{[k]}\|_2^2 \right)$$

$$\mathbf{z}^{[k+1]} = \arg \min_{\mathbf{z}} \left(g(\mathbf{z}) + (\rho/2) \|\mathbf{x}^{[k+1]} - \mathbf{z} - \lambda^{[k]}\|_2^2 \right)$$

$$\lambda^{[k+1]} = \lambda^{[k]} - \mathbf{x}^{[k+1]} + \mathbf{z}^{[k+1]}$$

- $\rho > 0$ is a step size; λ is the dual variable associated the constraint

Alternating direction method of multipliers

- **Convergence of ADMM:** Under benign conditions ADMM guarantees
 - $f(\mathbf{x}^k) + g(\mathbf{z}^k) \rightarrow p^\star$
 - $\lambda^k \rightarrow \lambda^\star$, an optimal dual variable
 - $\mathbf{x}^k - \mathbf{z}^k \rightarrow 0$
- Same as many other operator splitting methods for consensus problem, e.g., Douglas-Rachford method
- **Pros:** 1) with good robustness of method of multipliers; 2) can support decomposition

Operator splitting

- Transform HSD embedding $\mathcal{F}_{\text{HSD}}$ in ADMM form: Apply the operating splitting method (ADMM)

$$\begin{aligned} \mathcal{P}_{\text{ADMM}} : \underset{\mathbf{x}, \tilde{\mathbf{x}}, \mathbf{y}, \tilde{\mathbf{y}}}{\text{minimize}} \quad & I_{\mathcal{C} \times \mathcal{C}^*}(\mathbf{x}, \mathbf{y}) + I_{\mathbf{Q}\tilde{\mathbf{x}}=\tilde{\mathbf{y}}}(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \\ \text{subject to} \quad & (\mathbf{x}, \mathbf{y}) = (\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \end{aligned}$$

- **Final algorithm**

$$\begin{aligned} \tilde{\mathbf{x}}^{[i+1]} &= (\mathbf{I} + \mathbf{Q})^{-1}(\mathbf{x}^{[i]} + \mathbf{y}^{[i]}) && \text{subspace projection} \\ \mathbf{x}^{[i+1]} &= \Pi_{\mathcal{C}}(\tilde{\mathbf{x}}^{[i+1]} - \mathbf{y}^{[i]}) && \text{parallel cone projection} \\ \mathbf{y}^{[i+1]} &= \mathbf{y}^{[i]} - \tilde{\mathbf{x}}^{[i+1]} + \mathbf{x}^{[i+1]} && \text{computationally trivial} \end{aligned}$$

Parallel cone projection

- **Proximal algorithms** for parallel cone projection [Parikh & Boyd, FTO 14]

- Projection onto the second-order cone: $\mathcal{Q}^d = \{(z, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^{d-1} \mid \|\mathbf{x}\| \leq z\}$

$$\Pi_{\mathcal{C}}(\boldsymbol{\omega}, \tau) = \begin{cases} 0, & \|\boldsymbol{\omega}\|_2 \leq -\tau \\ (\boldsymbol{\omega}, \tau), & \|\boldsymbol{\omega}\|_2 \leq \tau \\ (1/2)(1 + \tau/\|\boldsymbol{\omega}\|_2)(\boldsymbol{\omega}, \|\boldsymbol{\omega}\|_2), & \|\boldsymbol{\omega}\|_2 \geq |\tau|. \end{cases}$$

- Closed-form, computationally scalable (we mainly focus on SOCP)
- Projection onto positive semidefinite cone: $\mathbf{S}_+^n = \{\mathbf{M} \in \mathbb{R}^{n \times n} \mid \mathbf{M} = \mathbf{M}^T, \mathbf{M} \succeq \mathbf{0}\}$

$$\Pi_{\mathcal{C}}(\mathbf{V}) = \sum_{i=1}^n (\lambda_i)_+ \mathbf{u}_i \mathbf{u}_i^T$$

- SVD is computationally expensive

Numerical results

- Power minimization coordinated beamforming problem

Network Size ($L=K$)		20	50	100	150
CVX+SDPT3	Modeling Time [sec]	0.7563	4.4301	N/A	N/A
	Solving Time [sec]	4.2835	326.2513	N/A	N/A
	Objective [W]	12.2488	6.5216	N/A	N/A
Matrix Stuffing+ADMM	Modeling Time [sec]	0.0128	0.2401	2.4154	9.4167
	Solving Time [sec]	0.1009	2.4821	23.8088	81.0023
	Objective [W]	12.2523	6.5193	3.1296	2.0689

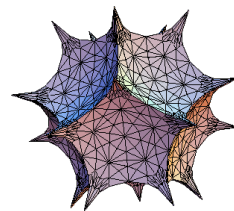
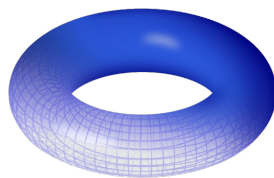
Matrix stuffing can speedup **60x** over CVX

ADMM can speedup **130x** over the interior-point method

[Ref] Y. Shi, J. Zhang, B. O'Donoghue, and K. B. Letaief, "Large-scale convex optimization for dense wireless cooperative networks," IEEE Trans. Signal Process., vol. 63, no. 18, pp. 4729-4743, Sept. 2015. **(The 2016 IEEE Signal Processing Society Young Author Best Paper Award)**

Vignette B: **Scalable Optimization on Manifolds**

1. Motivation: Why Nonconvex Optimization?
 - 1) Geometry of Nonconvex Optimization
2. Riemannian Optimization Algorithms



Optimization over Riemannian Manifolds (non-Euclidean geometry)

*Motivation: Why **Nonconvex** Optimization?*

Low-rank matrix optimization

- Rank-constrained matrix optimization problem

$$\underset{\mathbf{M} \in \mathbb{R}^{n \times n}}{\text{minimize}} \quad f(\mathcal{A}(\mathbf{M})) \quad \text{subject to} \quad \text{rank}(\mathbf{M}) = r$$

- $\mathcal{A} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^d$ is a real linear map on $n \times n$ matrices
- $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and differentiable
- A prevalent **model** in signal processing, statistics, and machine learning
- **Challenge I:** **Reliably** solve the low-rank matrix problem **at scale**
- **Challenge II:** Develop optimization algorithms with **optimal storage** $\Theta(rn)$

A brief biased history of convex methods

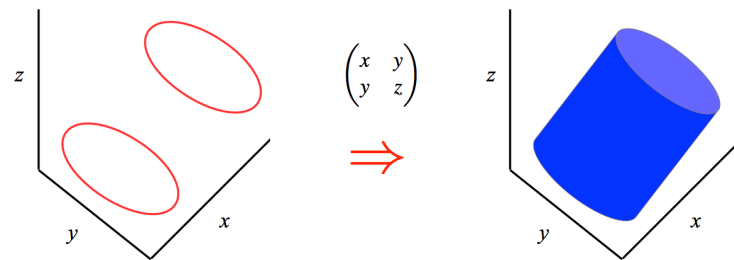
- **1990s:** Interior-point methods (**computationally expensive**)
 - **Storage cost** $\Theta(n^4)$ **for Hessian**
- **2000s:** Convex first-order methods
 - (Accelerated) proximal gradient, spectral bundle methods, and others
 - **Store matrix variable** $\Theta(n^2)$
- **2008-Present:** Storage-efficient convex first-order methods
 - Conditional gradient method (CGM) and extensions
 - Store matrix in low-rank form $\mathcal{O}(tn)$ after t iterations: **no storage guarantees**

Interior-point: Nemirovski & Nesterov 1994; ... **First-order:** Rockafellar 1976; Helmberg & Rendl 1997; Auslender & Teboulle 2006; ... **CGM:** Frank & Wolfe 1956; Levitin & Poljak 1967; Jaggi 2013; ...

Convexity: Why bother?

- **Convex relaxation fails:** always return the identity matrix!

$$\begin{aligned} & \underset{M \in \mathbb{C}^{K \times K}}{\text{minimize}} && \|M\|_* \\ & \text{subject to} && M_{ii} = 1, i = 1, \dots, K \\ & && M_{ij} = 0, \forall (i, j) \in \mathcal{S} \end{aligned}$$



- **Fact:** $\text{Trace}(M) \leq \|M\|_*$
- **The dilemma:** Convex methods have slow memory hogs, high computational complexity, sometimes fail

Can we solve the nonconvex matrix optimization problem directly?

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \|X\|_* \\ \text{subject to} & y = A(x) \end{array}$$

$\Downarrow$ SDP

$$\begin{array}{ll} \underset{X, W_1, W_2}{\text{minimize}} & \frac{1}{2} \text{Tr}(W_1) + \frac{1}{2} \text{Tr}(W_2) \\ \text{subject to} & y = A(x), \quad \underbrace{\begin{bmatrix} W_1 & X \\ X^T & W_2 \end{bmatrix}} \succeq 0 \end{array}$$

$$\text{lemma: } \|X\|_* = \min_{W_1, W_2} \left\{ \frac{1}{2} \text{Tr}(W_1) + \frac{1}{2} \text{Tr}(W_2) \mid \begin{bmatrix} W_1 & X \\ X^T & W_2 \end{bmatrix} \succeq 0 \right\}$$

$$\text{optimal point: } W_1 = U \Sigma U^T, \quad W_2 = V \Sigma V^T$$

(where $X = U \Sigma V^T$)

$$\|X\|_* = \sum_{i=1}^n b_i(x)$$

i^{th} largest singular value

Recent advances in nonconvex optimization

- **2009–Present: Nonconvex heuristics**

- Burer–Monteiro factorization idea + various nonlinear programming methods
- Store low-rank matrix factors $\Theta(rn)$

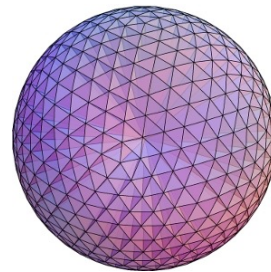
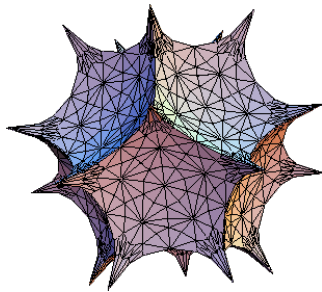
- **Guaranteed solutions:** Global optimality with statistical assumptions

- Matrix completion/recovery: [Sun-Luo'14], [Chen-Wainwright'15], [Ge-Lee-Ma'16],...
- Phase retrieval: [Candes et al., 15], [Chen-Candes'15], [Sun-Qu-Wright'16]
- Community detection/phase synchronization [Bandeira-Boumal-Voroninski'16], [Montanari et al., 17],...

When are nonconvex optimization problems not scary?

Riemannian Optimization Algorithms

Escape saddle points via manifold optimization

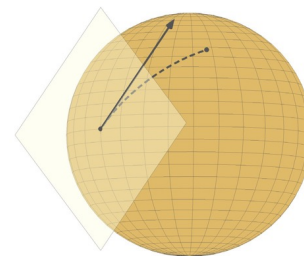


What is manifold optimization?

- Manifold (or manifold-constrained) optimization problem

$$\underset{M \in \mathbb{C}^{m \times n}}{\text{minimize}} \quad f(M) \quad \text{subject to} \quad M \in \mathcal{M}$$

- $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a **smooth function**
- $\mathcal{M}$ is a **Riemannian manifold**: **spheres**, orthonormal bases (Stiefel), rotations, **positive definite matrices**, **fixed-rank matrices**, Euclidean distance matrices, **semidefinite fixed-rank matrices**, **linear subspaces (Grassmann)**, **phases**, **essential matrices**, **fixed-rank tensors**, Euclidean spaces...



Escape saddle points via manifold optimization

- Convergence guarantees for Riemannian **trust regions**
 - Global convergence to **second-order critical points**
 - Quadratic convergence rate locally
 - Reach ϵ -**second order stationary point** $\|\text{grad} f(z)\| \leq \epsilon$ and $\nabla^2 f(z) \succeq -\epsilon I$ in $\mathcal{O}(1/\epsilon^3)$ iterations under Lipschitz assumptions [Cartis & Absil'16]

Escape saddle points via finding second-order stationary point

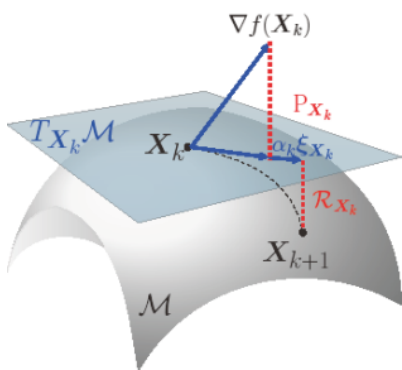
- **Other approaches:** Gradient descent by adding noise [Ge et al., 2015], [Jordan et al., 17] (slow convergence rate in general)

Recent applications of manifold optimization

- Matrix/tensor completion/recovery: [Vandereycken'13], [Boumal-Absil'15], [Kasai-Mishra'16],...
- Gaussian mixture models: [Hosseini-Sra'15], Dictionary learning: [Sun-Qu-Wright'17], Phase retrieval: [Sun-Qu-Wright'17],...
- Phase synchronization/community detection: [Boumal'16], [Bandeira-Boumal-Voroninski'16],...
- **Wireless transceivers design:** [Shi-Zhang-Letaief'16], [Yu-Shen-Zhang-K. B. Letaief'16], [Shi-Mishra-Chen'16],...

The power of manifold optimization paradigms

- Generalize Euclidean gradient (Hessian) to *Riemannian gradient (Hessian)*



$$\nabla_{\mathcal{M}} f(\mathbf{X}^{(k)}) = P_{\mathbf{X}^{(k)}}(\nabla f(\mathbf{X}^{(k)}))$$

Riemannian Gradient Euclidean Gradient

$$\mathbf{X}^{(k+1)} = \mathcal{R}_{\mathbf{X}^{(k)}}(-\alpha^{(k)} \nabla_{\mathcal{M}} f(\mathbf{X}^{(k)}))$$

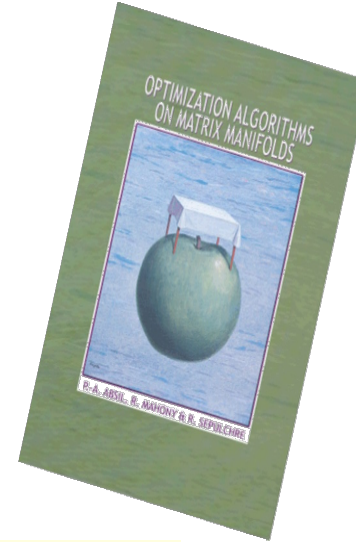
Retraction Operator

- We need Riemannian geometry: 1) linearize search space $\mathcal{M}$ into a **tangent space** $T_{\mathbf{X}}\mathcal{M}$; 2) pick a **metric** on $T_{\mathbf{X}}\mathcal{M}$ to give intrinsic notions of **gradient** and **Hessian**

An excellent book

Optimization algorithms on matrix manifolds

A Matlab toolbox



[Manopt](#) [Home](#) [Tutorial](#) [Forum](#) [About](#) [Contact](#)

Welcome to Manopt!

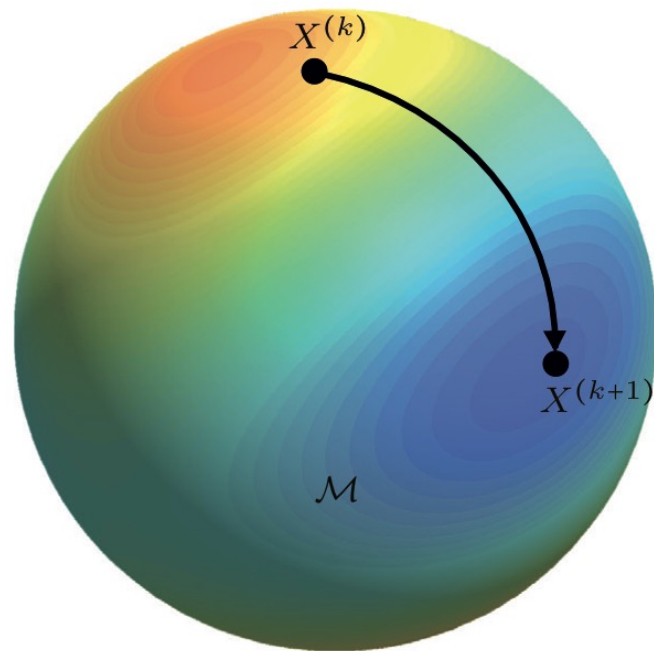
A Matlab toolbox for optimization on manifolds

Optimization on manifolds is a powerful paradigm to address nonlinear optimization problems. With Manopt, it is easy to deal with various types of constraints that arise naturally in applications, such as orthonormality or low rank.

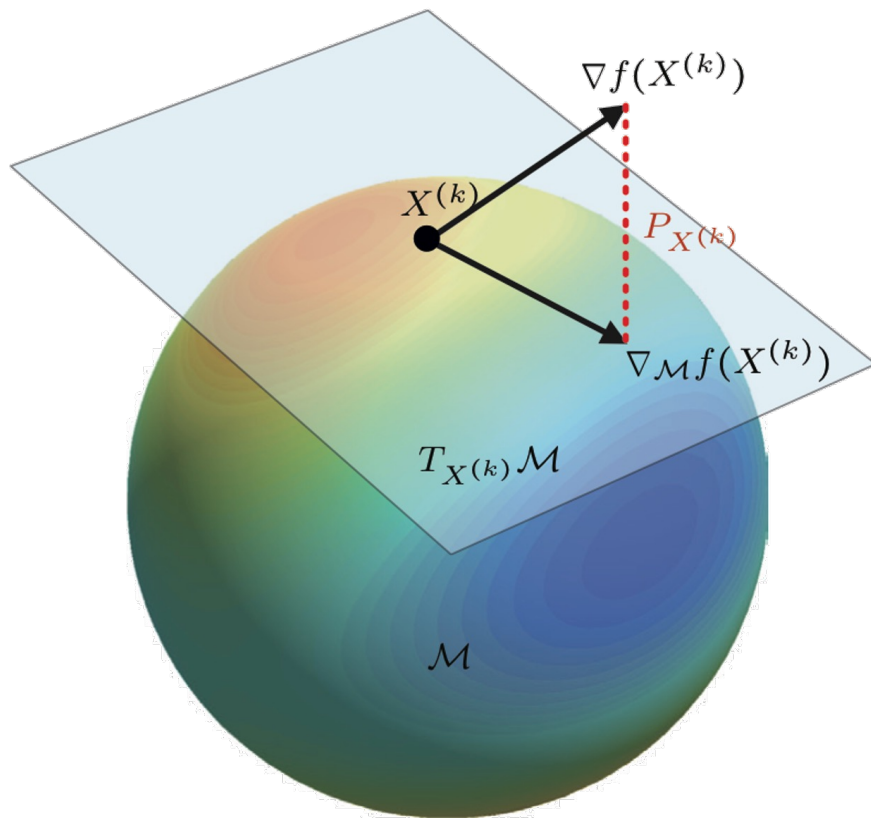
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*Taking A Close Look at **Gradient Descent***

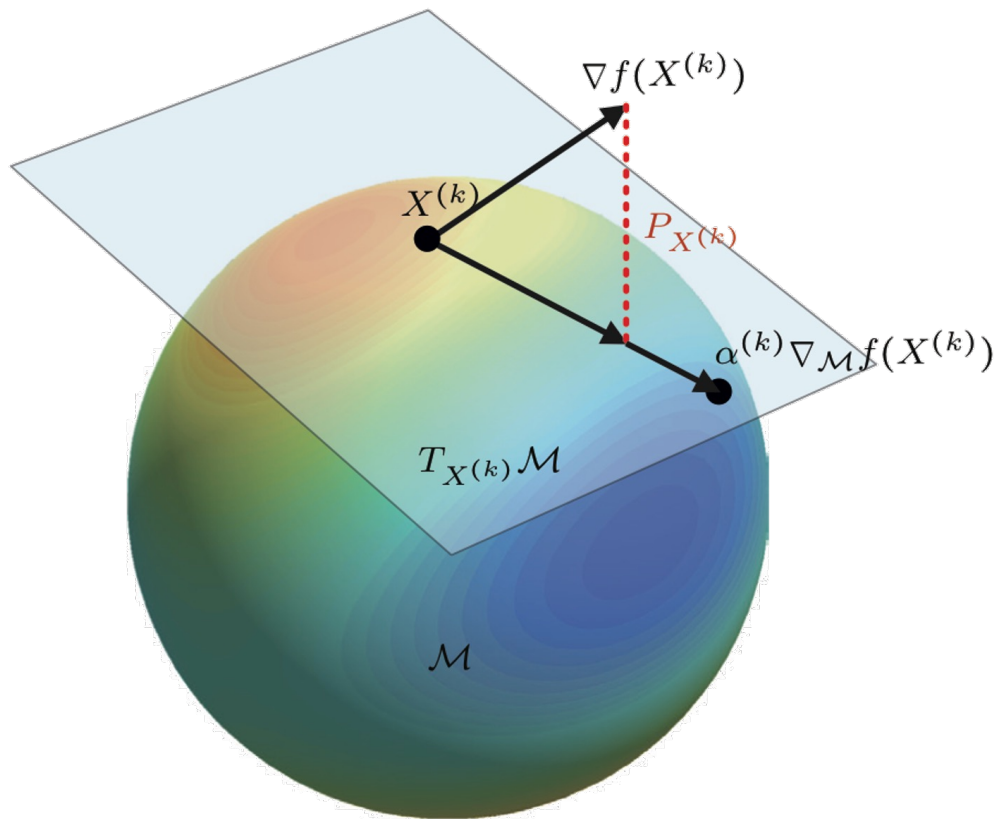
Optimization on the manifold: main idea



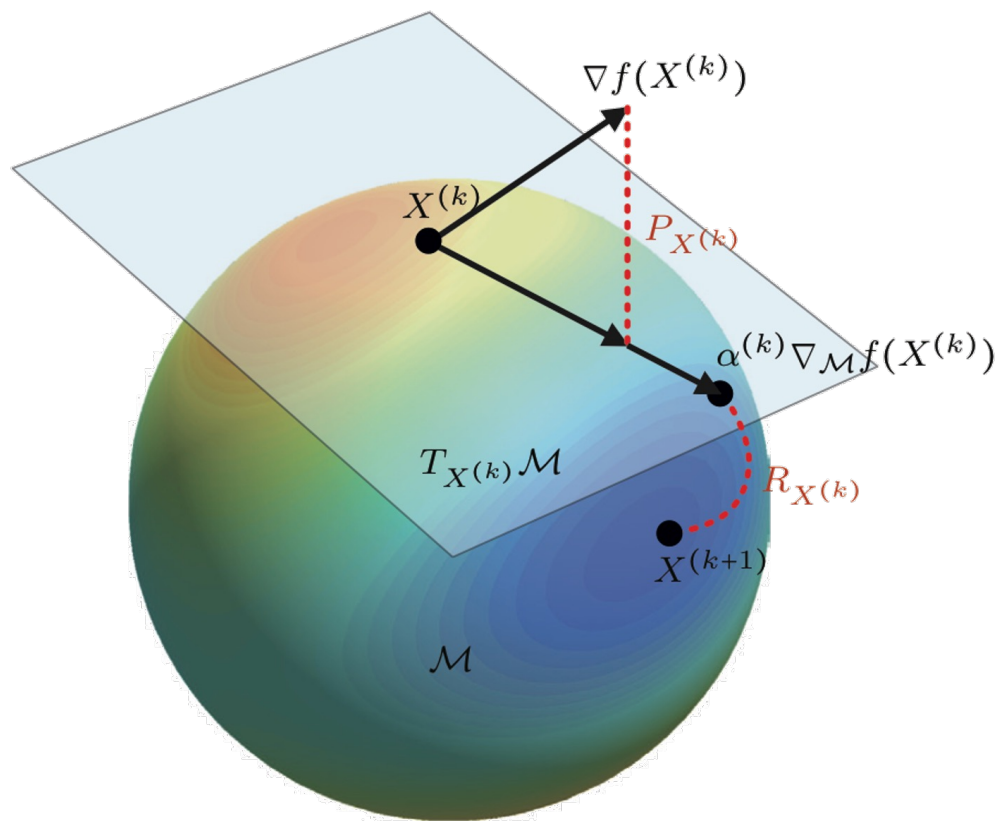
Optimization on the manifold: main idea



Optimization on the manifold: main idea



Optimization on the manifold: main idea



Example: Rayleigh quotient

- Optimization over (sphere) manifold $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : x^T x = 1\}$

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x) = -x^T A x \quad \text{subject to} \quad x^T x = 1$$

- The cost function is smooth on $\mathbb{S}^{n-1}$, symmetric matrix $A \in \mathbb{R}^{n \times n}$
- Step 1: Compute the **Euclidean gradient** in $\mathbb{R}^n$
$$\nabla f(x) = -2Ax$$
- Step 2: Compute the **Riemannian gradient** on $\mathbb{S}^{n-1}$ via projecting $\nabla f(x)$ to the tangent space using the orthogonal projector $\text{Proj}_x u = (I - xx^T)u$

$$\text{grad} f(x) = \text{Proj}_x \nabla f(x) = -2(I - xx^T)Ax$$

Example: Generalized low-rank optimization

- Generalized low-rank optimization for topological interference alignment via Riemannian optimization

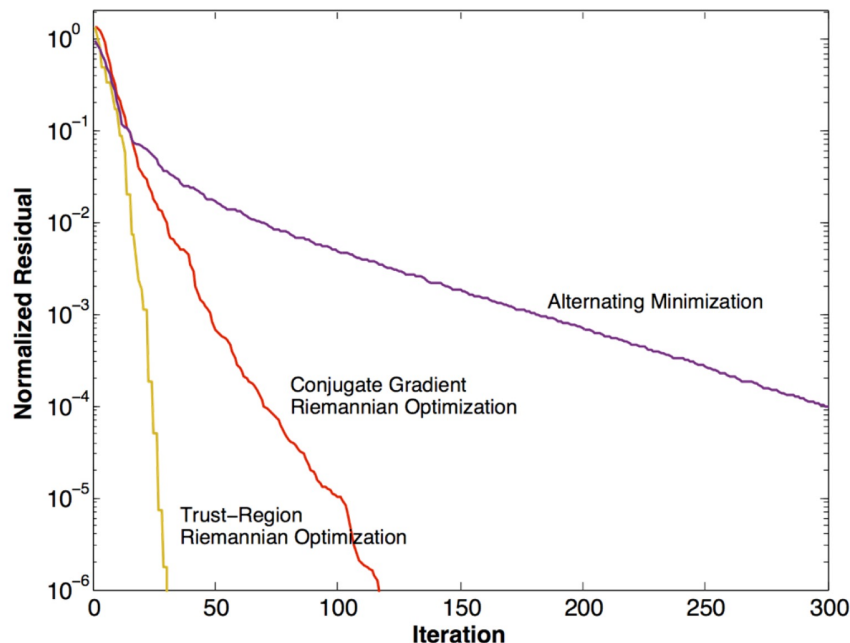
$$\underset{\mathbf{M} \in \mathbb{C}^{m \times n}}{\text{minimize}} \quad f(\mathbf{M}), \quad \text{subject to} \quad \text{rank}(\mathbf{M}) = r$$

OPTIMIZATION-RELATED INGREDIENTS FOR PROBLEM $\mathcal{P}_r$

	$\mathcal{P}_r : \underset{\mathbf{X} \in \mathcal{M}_r}{\text{minimize}} f(\mathbf{X})$
Matrix representation of an element $\mathbf{X} \in \mathcal{M}_r$	$\mathbf{X} = (\mathbf{U}, \mathbf{\Sigma}, \mathbf{V})$
Computational space $\mathcal{M}_r$	$\text{St}(r, M) \times \text{GL}(r) \times \text{St}(r, M)$
Quotient space	$\text{St}(r, M) \times \text{GL}(r) \times \text{St}(r, M) / (\mathcal{O}(r) \times \mathcal{O}(r))$
Metric $g_{\mathbf{X}}(\xi_{\mathbf{X}}, \zeta_{\mathbf{X}})$ for $\xi_{\mathbf{X}}, \zeta_{\mathbf{X}} \in T_{\mathbf{X}}\mathcal{M}_r$	$g_{\mathbf{X}}(\xi_{\mathbf{X}}, \zeta_{\mathbf{X}}) = \langle \xi_U, \zeta_U \Sigma \Sigma^T \rangle + \langle \xi_{\Sigma}, \zeta_{\Sigma} \rangle + \langle \xi_V, \zeta_V \Sigma^T \Sigma \rangle$
Riemannian gradient $\text{grad}_{\mathbf{X}} f$	$\text{grad}_{\mathbf{X}} f = (\xi_U, \xi_{\Sigma}, \xi_V) \quad (30)$
Riemannian Hessian $\text{Hess}_{\mathbf{X}} f[\xi_{\mathbf{X}}]$	$\text{Hess}_{\mathbf{X}} f[\xi_{\mathbf{X}}] = \Pi_{\mathcal{H}_{\mathbf{X}}\mathcal{M}_r} (\nabla_{\xi_{\mathbf{X}}} \text{grad}_{\mathbf{X}} f) \quad (40)$
Retraction $\mathcal{R}_{\mathbf{X}}(\xi_{\mathbf{X}}) : \mathcal{H}_{\mathbf{X}}\mathcal{M}_r \rightarrow \mathcal{M}_r$	$(\text{uf}(\mathbf{U} + \xi_U), \mathbf{\Sigma} + \xi_{\Sigma}, \text{uf}(\mathbf{V} + \xi_V))$

Convergence rates

- Optimize over fixed-rank matrices (quotient matrix manifold)



Riemannian algorithms:

1. Exploit the rank structure in a principled way
2. Develop second-order algorithms systematically
3. Scalable, SVD-free

[Ref] Y. Shi, J. Zhang, and K. B. Letaief, “Low-rank matrix completion for topological interference management by Riemannian pursuit,” *IEEE Trans. Wireless Commun.*, vol. 15, no. 7, Jul. 2016.

Concluding remarks

- **Large-scale convex optimization**

- Convex geometry and analysis provide optimality guarantees
- Matrix stuffing for fast HSD embedding transformation
- Operator splitting for solving large-scale HSD embedding

- **Future directions:**

- Optimality guarantees for more complicated problems, e.g., group sparse beamforming
- Operator splitting for general large-scale SDP problems, e.g., using approximated cone projection
- More applications: deep neural network compression via sparse optimization

Concluding remarks

- **Scalable nonconvex optimization algorithms**
 - Nonconvex statistical optimization may not be that scary: no spurious local optima
 - Riemannian optimization is powerful: 1) Exploit the manifold geometry of fixed-rank matrices; 2) Escape saddle points
- **Future directions:**
 - Geometry of neural network loss surfaces: saddle points, local/global optima
 - More applications: blind deconvolution for IoT, big data analytics (e.g., ranking)

To learn more...

- **Web:** <http://shiyuanming.github.io/sparserank.html>
- **Papers:**
- Y. Shi, J. Zhang, and K. B. Letaief, “Group sparse beamforming for green Cloud-RAN,” *IEEE Trans. Wireless Commun.*, vol. 13, no. 5, pp. 2809-2823, May 2014. (The 2016 Marconi Prize Paper Award)
- Y. Shi, J. Zhang, B. O’Donoghue, and K. B. Letaief, “Large-scale convex optimization for dense wireless cooperative networks,” *IEEE Trans. Signal Process.*, vol. 63, no. 18, pp. 4729-4743, Sept. 2015. t. 2015. (The 2016 IEEE Signal Processing Society Young Author Best Paper Award)
- Y. Shi, J. Zhang, K. B. Letaief, B. Bai and W. Chen, “Large-scale convex optimization for ultra-dense Cloud-RAN,” *IEEE Wireless Commun. Mag.*, pp. 84-91, Jun. 2015.
- Y. Shi, J. Zhang, W. Chen, and K. B. Letaief, “Generalized sparse and low-rank optimization for ultra-dense networks,” *IEEE Commun. Mag.*, vol. 56, no. 6, pp. 42-48, Jun., 2018.

To learn more...

- Y. Shi, J. Zhang, and K. B. Letaief, “Optimal stochastic coordinated beamforming for wireless cooperative networks with CSI uncertainty,” *IEEE Trans. Signal Process.*, vol. 63, no. 4, pp. 960-973, Feb. 2015.
- Y. Shi, J. Zhang, and K. B. Letaief, “Robust group sparse beamforming for multicast green Cloud- RAN with imperfect CSI,” *IEEE Trans. Signal Process.*, vol. 63, no. 17, pp. 4647-4659, Sept. 2015.
- Y. Shi, J. Cheng, J. Zhang, B. Bai, W. Chen and K. B. Letaief, “Smoothed L_p -minimization for green Cloud-RAN with user admission control,” *IEEE J. Select. Areas Commun.*, vol. 34, no. 4, Apr. 2016.
- X. Yu, J.-C. Shen, J. Zhang, and K. B. Letaief, “Alternating minimization algorithms for hybrid precoding in millimeter wave MIMO systems,” *IEEE J. Sel. Topics Signal Process.*, vol. 10, no. 3, pp. 485-500, Apr. 2016.
- Y. Shi, J. Zhang, and K. B. Letaief, “Low-rank matrix completion for topological interference management by Riemannian pursuit,” *IEEE Trans. Wireless Commun.*, vol. 15, no. 7, Jul. 2016.
- Y. Shi, B. Mishra, and W. Chen, “Topological interference management with user admission control via Riemannian optimization,” *IEEE Trans. Wireless Commun.*, vol. 16, no. 11, pp. 7362-7375, Nov. 2017.
- X. Peng, Y. Shi, J. Zhang, and K. B. Letaief, “Layered group sparse beamforming for cache-enabled wireless networks,” *IEEE Trans. Commun.*, vol. 65, no. 12, pp. 5589-5603, Nov. 2017.
- Y. Shi, J. Zhang, W. Chen, and K. B. Letaief, “Enhanced group sparse beamforming for dense green Cloud-RAN: A random matrix approach,” *IEEE Trans. Wireless Commun.*, vol. 17, no. 4, pp. 2511-2524, Apr. 2018.
- K. Yang, Y. Shi, and Z. Ding, “Generalized low-rank optimization for topological cooperation in ultra-dense networks,” *IEEE Trans. Wireless Commun.*, to appear, 2019.